Gradient-Boosted Trees and Consensus-Risk

For Opioid and Polypharmacy Emergency Department Visits in Virginia's APCD

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Roadmap

FINAL DEFENSE AGENDA | 7 CORE MODULES

I. Problem

- Dual Crises
- Gap
- Aims

II. Governance & Data

- CRISP-DM
- Virginia APCD

III. Cohorts

- Opioid (12m)
- Polypharm (21d)

IV. Modeling

- GBDT
- SHAP ∩ FFA
- Step 3b

V. Translation & Deployment

- OODA Loop
- Study Fences

VI. Results

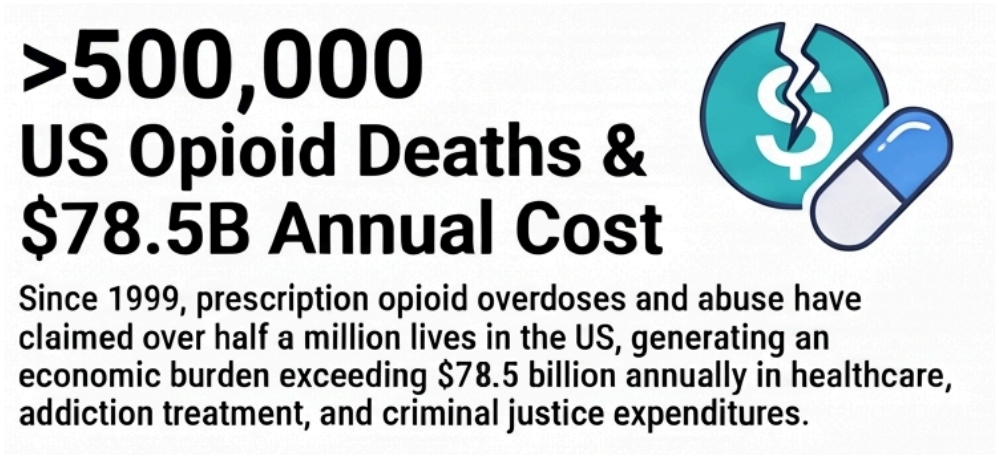
- CH2 Data Eng
- CH3 Opioid
- CH4 Polypharm
- CH5 CDS

VII. Next Steps

- Pending 9NOV Feedback
- CCTR & SaMD Horizon

Downstream Clinical & Economic Effects

MODULE I | INFOGRAPHIC PART 1 OF 4 · DOWNSTREAM PUBLIC HEALTH & ECONOMIC TOLL



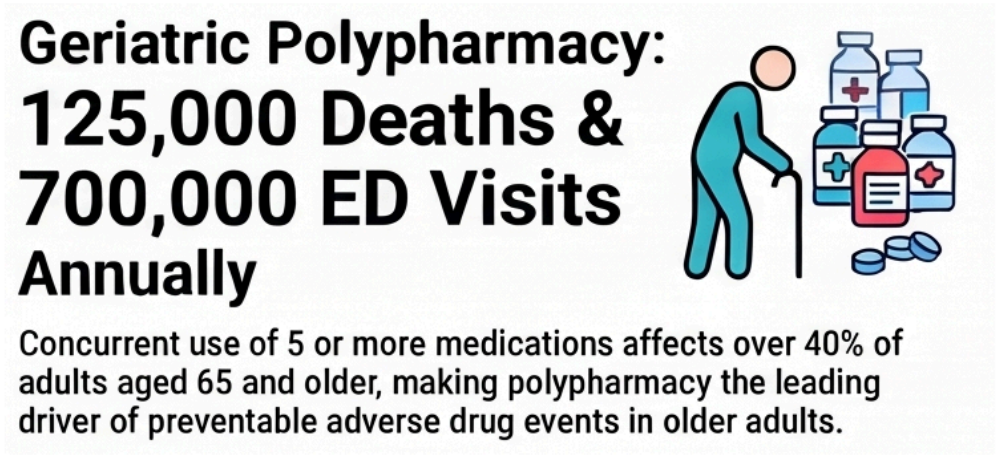
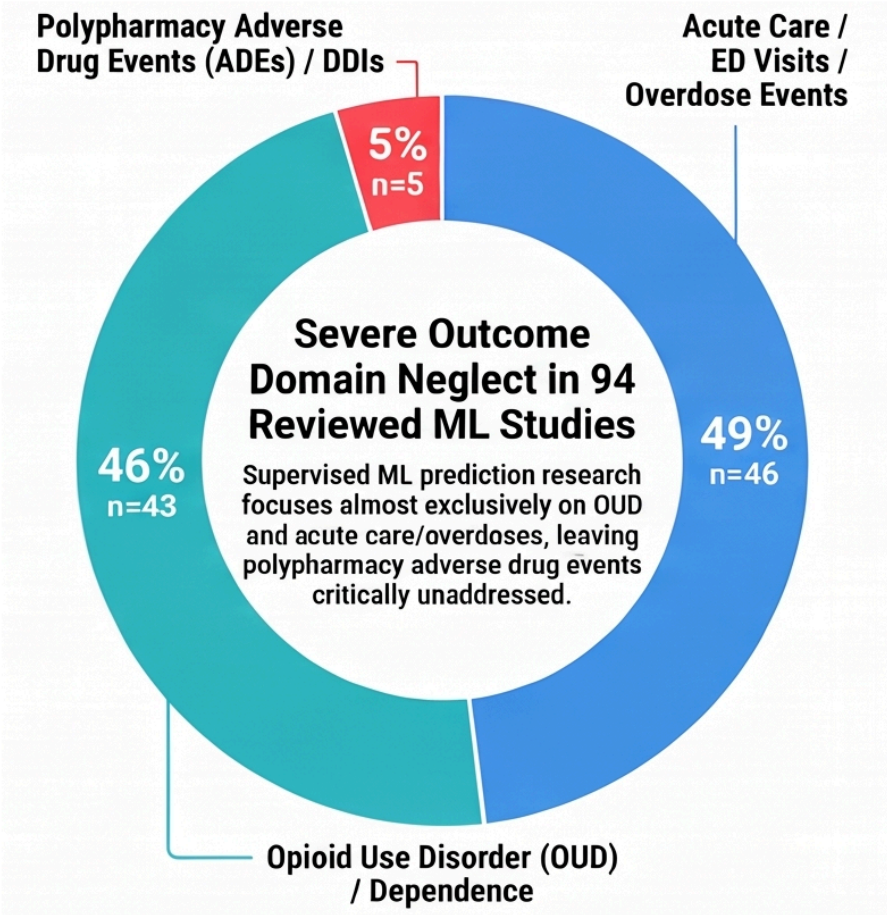


Figure 1A (Part 1): Downstream clinical and economic effects of US opioid overdoses and geriatric polypharmacy (CDC WONDER; Florence et al., *Med Care*; Charlesworth et al., *J Gerontol A*; Budnitz/Shehab et al., *JAMA*).

Opioid Research Overshadows Polypharmacy

MODULE I | INFOGRAPHIC PART 2 OF 4 · SYSTEMATIC REVIEW (N = 94 ML STUDIES)



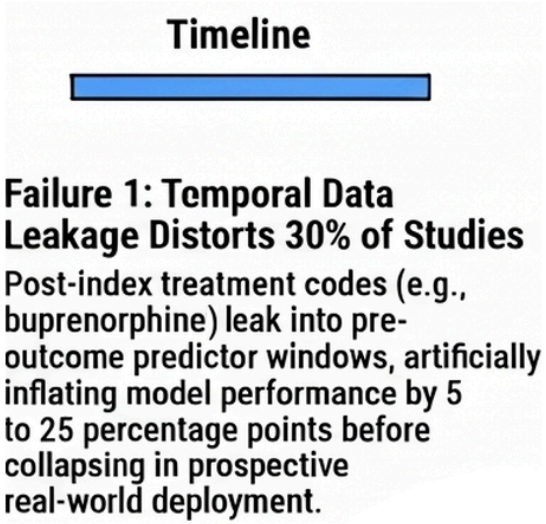
Claims-Based Opioid ED Modeling Void

Out of 1,473 published OUD literature records, only 6 studies specifically evaluate machine learning models designed to predict opioid-related emergency department visits using administrative claims data.

Figure 1A (Part 2): Published outcome domain distribution across 94 supervised ML studies and the claims-based opioid ED modeling void.

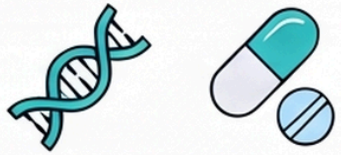
ML Gap – No PGx features. No XAI. No Target Leakage Validation.

MODULE I | INFOGRAPHIC PART 3 OF 4 · PROBAST APPRAISAL (N = 73 MODELS)



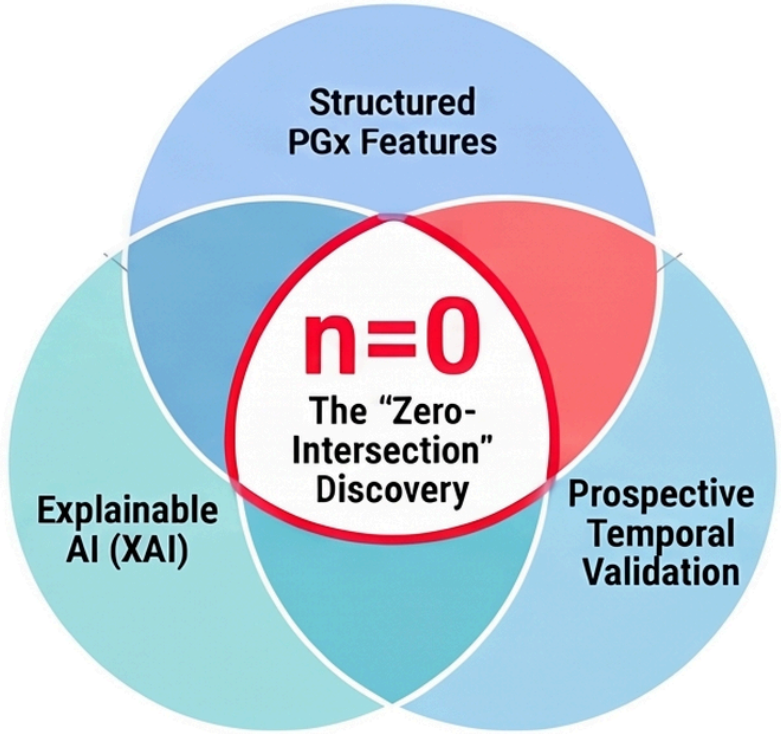
Failure 1: Temporal Data Leakage Distorts 30% of Studies

Post-index treatment codes (e.g., buprenorphine) leak into pre-outcome predictor windows, artificially inflating model performance by 5 to 25 percentage points before collapsing in prospective real-world deployment.



Failure 2: 81% Omit Pharmacogenomics (PGx)

Despite mature CPIC guidelines and known gene-drug modifiers (e.g., CYP2D6, 0PRM1), 81% of reviewed ML prediction pipelines completely exclude structured pharmacogenomic features.



Across 73 PROBAST-assessed modeling studies, zero published systems successfully combine structured PGx features, explainable AI (XAI), and prospective temporal validation.

Figure 1A (Part 3): PROBAST appraisal of 73 predictive models exposing the ML gap ($n = 0$ combining PGx features, XAI, and target leakage / temporal holdout validation).

90% of ADEs Occur Within 21 Days

MODULE I | STEP 1

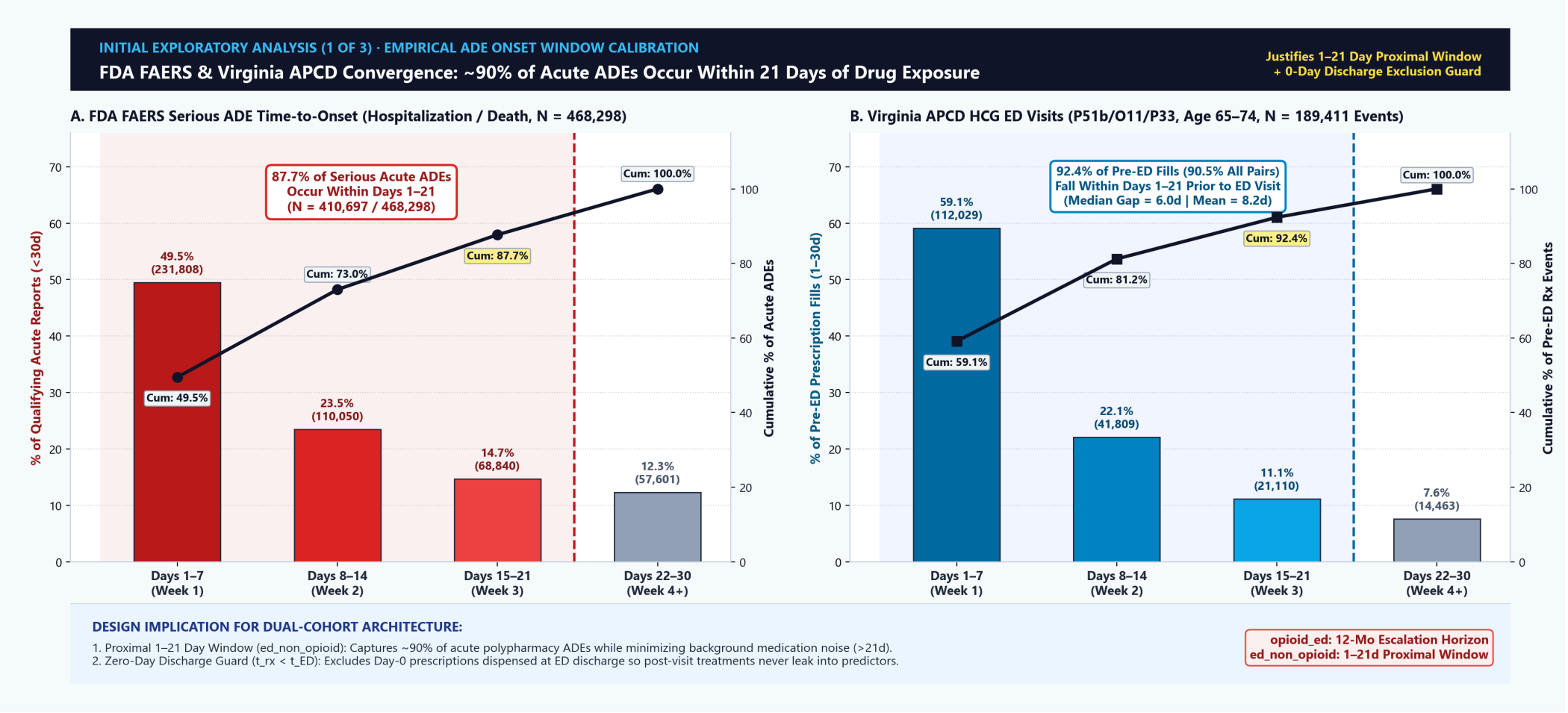


Figure 1B (Part 1): Convergent time-to-onset distribution across FDA FAERS ($N = 468,298$, 87.7% $\leq$ 21d) and Virginia APCD HCG ED visits ($N = 189,411$, 92.4% $\leq$ 21d).

Initial Feature Analysis

MODULE I | STEP 3A

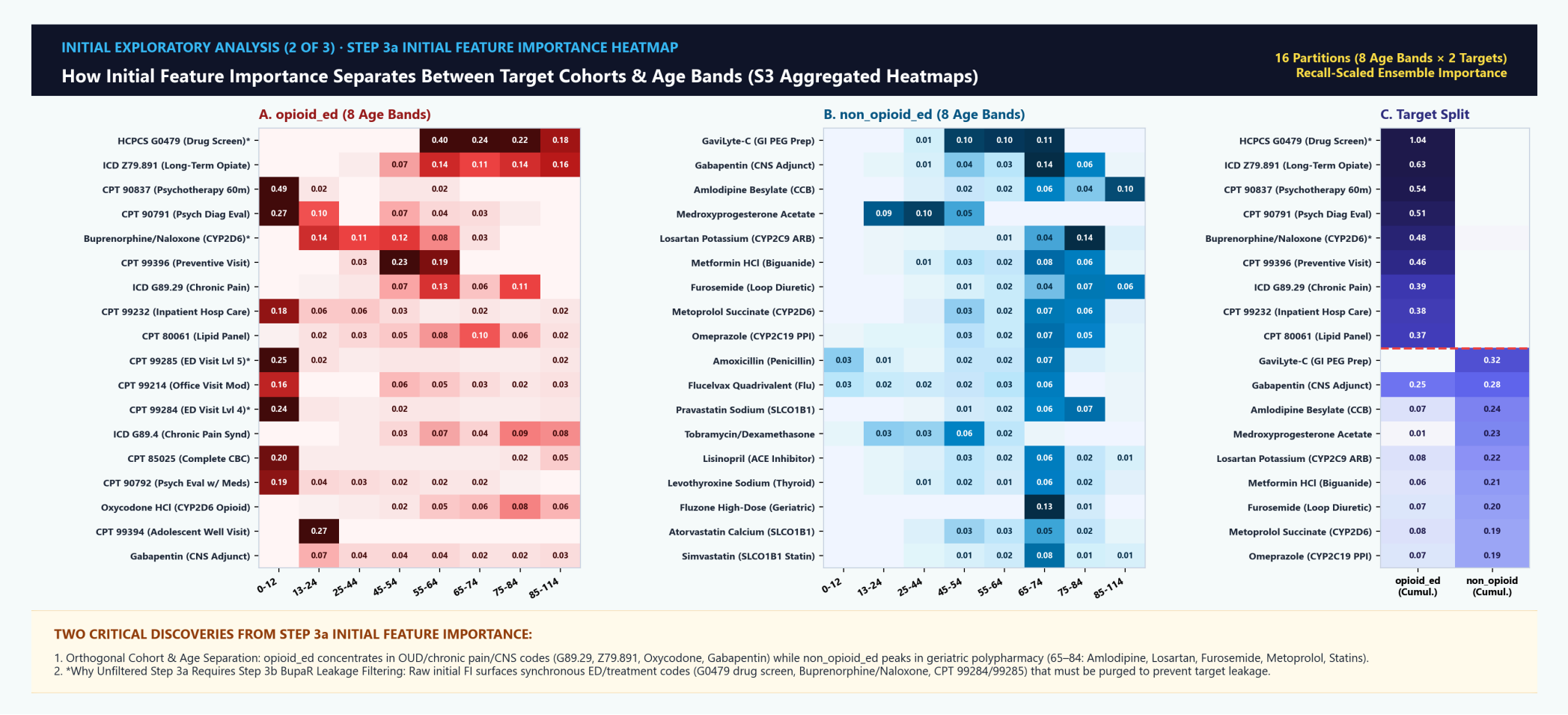


Figure 1B (Part 2): Step 3a multi-model (`CatBoost` , `XGBoost` , `XGBoost-RF`) aggregated feature importance across all 8 age bands (`0-12` to `85-114`) and both targets (`opioid_ed` vs. `non_opioid_ed`).

CPIC Drugs Drive Initial Feature Importance

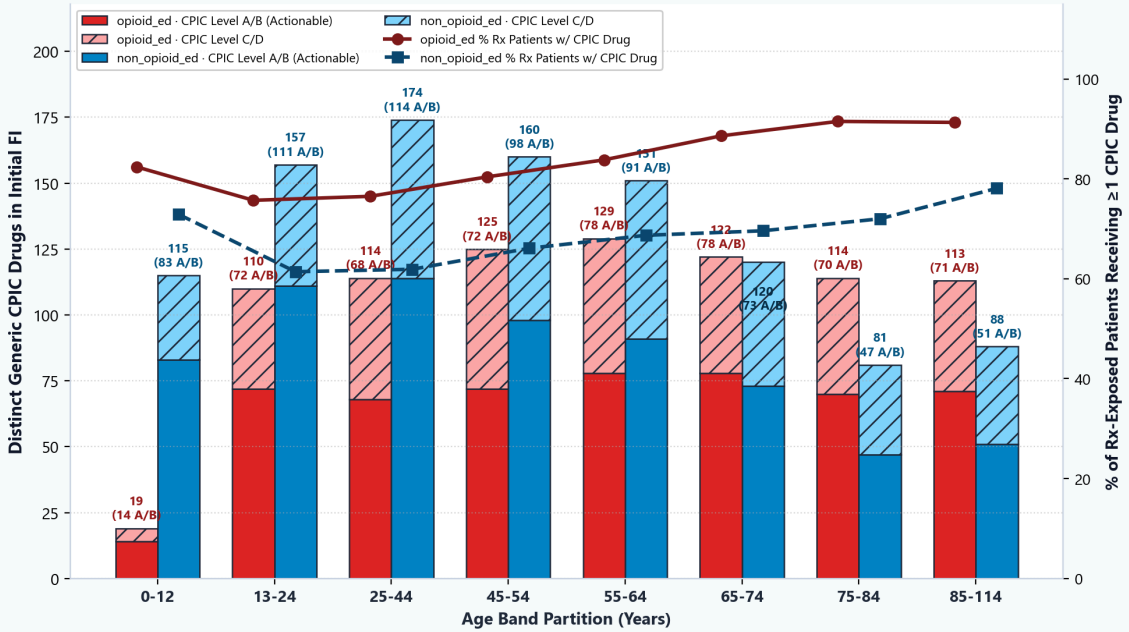
MODULE I | STEP 5

INITIAL EXPLORATORY ANALYSIS (3 OF 3) · CPIC PHARMACOGENOMIC DRUGS IN INITIAL FEATURE IMPORTANCE

Number of CPIC-Actionable Drugs Surfaced in Initial Feature Importance by Target Cohort & Age Band

opioid_ed: 187 Distinct CPIC Drugs (333 Features)
non_opioid_ed: 224 Distinct CPIC Drugs (439 Features)

A. Distinct CPIC Drugs in Initial Feature Importance by Target Cohort & Age Band



B. Dominant CPIC Gene–Drug Pairs in Initial Feature Importance by Target

Pharmacogene	CPIC Lvl	opioid_ed Top FI Drugs (187 CPIC Drugs | 76–92% Rx Pts)	non_opioid_ed Top FI Drugs (224 CPIC Drugs | 61–78% Rx Pts)
CYP2D6 (Analgesics, SSRIs, β-Blockers)	Level A / B	• Oxycodone (#17), Hydrocodone (#14) • Tramadol (#39), Codeine, Ondansetron • Venlafaxine, Risperidone, Aripiprazole	• Tramadol (#3), Hydrocodone (#6) • Metoprolol (#8), Carvedilol, Duloxetine • Ondansetron, Amitriptyline, Mirtazapine
CYP2C19 (PPIs, SSRIs, Antiplatelets)	Level A	• Omeprazole (8/8 bands), Pantoprazole • Sertraline, Citalopram, Escitalopram • Diazepam (8/8 bands), Amitriptyline	• Omeprazole (#9), Pantoprazole (#12) • Escitalopram (#21), Sertraline • Citalopram, Clopidogrel
CYP2C9 & VKORC1 (NSAIDs, Sulfonyl-ureas, Warfarin)	Level A / B	• Ibuprofen (8/8 bands), Meloxicam • Celecoxib, Naproxen, Diclofenac • Phenytoin (CYP2C9 / HLA-B)	• Losartan (#5), Diclofenac, Ibuprofen • Meloxicam, Celecoxib, Glimepiride • Warfarin (CYP2C9 / VKORC1)
SLCO1B1 & ABCG2 (Statins / Hepatic Transporters)	Level A	• Atorvastatin, Simvastatin • Pravastatin, Rosuvastatin • Co-prescribed in ages 45–85+	• Pravastatin (#12), Atorvastatin (#17) • Simvastatin (#18), Rosuvastatin • High geriatric polypharmacy importance
OPRM1 / COMT / UGT2B15 / G6PD	Level B / C	• Buprenorphine/Naloxone (#5) • Fentanyl (#24), Lorazepam (8/8 bands) • Sulfamethoxazole/TMP (8/8 bands)	• Lorazepam (#14), Ofloxacin (#5) • Ciprofloxacin, Sulfamethoxazole/TMP • Nitrofurantoin (G6PD)

EMPIRICAL BRIDGE TO STEP 5 PGx FEATURE ENGINEERING (pgx_num_drugs & pgx_num_cpic_drugs):

1. High CPIC Density in Top Predictors: 24%–35% of all drug features in initial Feature Importance are CPIC-actionable medications (187 in opioid_ed; 224 in non_opioid_ed).
2. Calibrated Scientific Framing: Claims capture CPIC-actionable medication exposure burden (identifying patients whose regimen warrants PGx review), NOT patient genotypes.

pgx_num_drugs & pgx_num_cpic_drugs
Rank #1–#3 Across Final Models

Figure 1B (Part 3): Number of distinct CPIC-actionable drugs in initial Feature Importance results by target cohort (`opioid_ed` vs. `non_opioid_ed`) and age band.

The Frontline Clinical Bottleneck

MODULE I | INFOGRAPHIC PART 4 OF 4 · POINT-OF-CARE TRANSLATION GAP

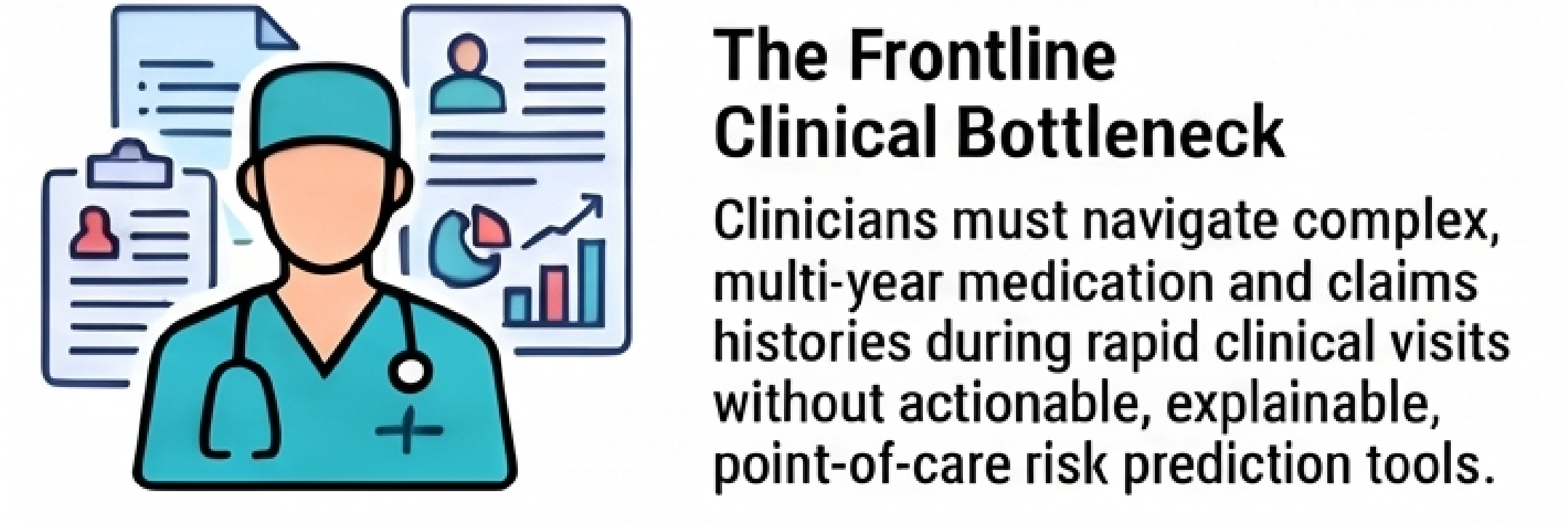


Figure 1A (Part 4): The Frontline & Predictive Clinical Bottleneck across longitudinal medication and administrative claims histories.

AIMS

MODULE I | AIM 1 · COHORT 1 (`opioid_ed`) | AIM 2 · COHORT 2 (`ed_non_opioid`) | AIM 3 · DECISION SUPPORT

Aim 1 · Cohort 1 Opioid ED Prediction & Attribution (Ch. 3)

Develop and temporally validate `CatBoost`, `XGBoost`, and `XGBoost-RF` models with $\mathrm{SHAP} \cap \mathrm{FFA}$ consensus attribution for **Cohort 1** (`opioid_ed` , 12-mo lookback, ages 13–64), evaluating pre-index leakage prevention, discrimination (`PR-AUC` , `AUROC`), and calibration on an independent **2019 temporal holdout**.

Aim 2 · Cohort 2 Polypharmacy & DDI Attribution (Ch. 4)

Develop and temporally validate `CatBoost`, `XGBoost`, and `XGBoost-RF` models with $\mathrm{SHAP} \cap \mathrm{FFA}$ Boolean Horn rules for **Cohort 2** (`ed_non_opioid` , 21-day lookback, ages 65–114) to identify reproducible risk predictors, candidate multi-drug interactions (`IE`), and consensus risk attenuation priorities (`IR`) on the **2019 holdout**.

Aim 3 · Usable Decision-Support Tool (Ch. 5)

Translate validated Aim 1 and Aim 2 risk estimates and interpretable Boolean rules into a usable prototype decision-support tool for pharmacist-led Medication Therapy Management (MTM) review prioritization—evaluating explanation fidelity, clinical interpretability, and workflow usability rather than claiming clinical effectiveness.

Cohort 1: Opioid-Related ED Events (`opioid_ed` , Ages 13–64)

- **Pathophysiology:** Progressive μ-opioid receptor tolerance, escalating MME titration, and synergistic CNS depressants (benzodiazepines, gabapentinoids) blunting respiratory drive.
- **Lookback Horizon: 12-Month Pre-Index Window** capturing chronic prescription escalation and behavioral comorbidities.
- **Claims Anchor (t_0):** Direct diagnostic billing for `F11.20` (opioid dependence) and `T40.0X` – `T40.4X` (opioid poisonings) scanned across **all 10 claim diagnostic positions**.

Cohort 2: Geriatric Polypharmacy & "Silent ADEs" (`ed_non_opioid` , Ages 65–114)

- **Why Pairwise CDS Fails:** Pairwise DDI ($A + B$) checkers trigger **90%–95% alert fatigue** while missing higher-order ($A + B + C + D$) synergies and age-related renal/hepatic decline.
- **The "Silent ADE" Coding Reality:** Fewer than **5%** of drug-induced ED visits are billed with explicit poisoning codes (`T36` – `T50`). Hospitals bill the **downstream clinical syndrome**:
  - **Falls & Syncope:** `R29.6` (falls), `S72` (hip fracture), `R55` (syncope) ← *Z-drugs, benzos, α-blockers*.
  - **Delirium, Bleeds & AKI:** `F05` (delirium), `K92.2` (GI bleed), `N17` (Triple-Whammy AKI).
- **Claims Anchor (t_0):** Milliman HCG emergency lines (`P51b` , `O11` , `P33`) anchored to a **21-Day Proximal Rx Window** (`0-day` guard), with **0 opioid codes** in study year.

Methodological Demarcation: Modeling Cohort 1 (12-month chronic opioid trajectory) and Cohort 2 (21-day proximal geriatric polypharmacy) separately prevents conflating distinct pathophysiologies, age distributions, and coding mechanics.

Governance and Frameworks

MODULE II | PHASE 1 · CRISP-DM, TRIPOD+AI & CLINICAL OODA LOOP

1. CRISP-DM (Systems Lifecycle)

- 6 iterative engineering phases governing the 10-step APCD pipeline.
- Ingestion → Cohort → Whitelist → Model → Eval → Microservice.

2. TRIPOD+AI (Biomedical Validation)

- Transparent reporting; Type 2b / Type 3 temporal validation.
- 2016–2018 train, untouched 2019 calendar holdout (zero post-hoc recalibration).

3. Clinical OODA Loop (Translation)

- **Observe / Orient:** GBT models & formal Boolean Horn rules.
- **Decide / Act:** Decision Support Tool

Appendix A6: Master Acronym & Framework Glossary (CRISP-DM, TRIPOD+AI, OODA) →

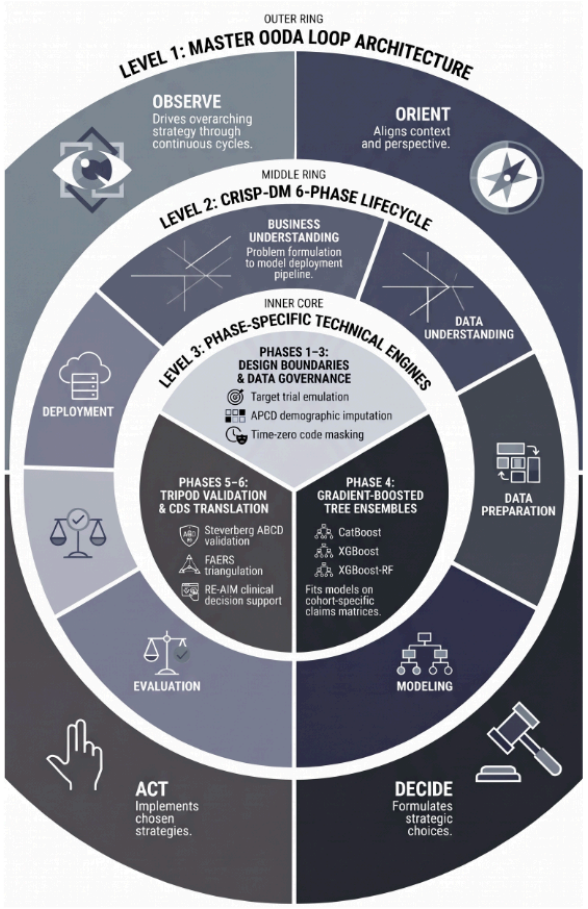


Figure 2: Three-Tier Governance Architecture: Master OODA Loop enclosing 6-phase CRISP-DM and phase-specific technical validation engines.

Why Virginia APCD Over EHR

MODULE II | PHASE 1 · STATEWIDE GENERALIZABILITY & DATA ARCHITECTURE

Virginia APCD (Selected)	**Single-System EHR** (Rejected)
Statewide Capture: Follows patients across health systems (eliminating out-of-network ED blindness).	**Fragmentation:** High loss-to-follow-up when crossing systems. Misclassifies out-of-network overdoses as "healthy non-events."
Medication Reality: Captures adjudicated pharmacy fills, proving physical drug possession.	**Intent vs Exposure:** Captures written prescriptions, missing ~30% primary non-adherence drop-offs.
Linkage: Single honest broker (VHI) guarantees zero probabilistic linkage attrition.	

The Methodological Compensation

We acknowledge the lack of molecular genotypes and real-time vitals in claims by computing CPIC PGx-burden proxies (*CYP2D6/CYP2C19* substrate counts) directly from longitudinal co-prescriptions.

Figure 3: Data Source Architecture: Statewide capture and adjudicated dispenses in Virginia APCD overcome single-system EHR fragmentation and prescription abandonment.

Why Virginia APCD Is Selected

- **Statewide Capture:** Follows patients across health systems, eliminating out-of-network ED blindness.
- **Medication Reality:** Captures paid/adjudicated dispenses, proving physical drug possession.
- **Linkage Integrity:** Single trusted data steward (VHI) guarantees 0 probabilistic linkage attrition.

The Methodological Compensation

- **EHR Genotype Gap:** APCD lacks molecular genotypes and real-time bedside vitals.
- **CPIC Guideline Mapping & Drug Counts:** We compensate by mapping dispensed APCD drugs to whether they carry a CPIC guideline (`drug_cpic_mapping_global.csv`) and counting patient-level total vs. CPIC drugs (`pgx_num_drugs` , `pgx_num_cpic_drugs`) and how many patients carry CPIC drugs.

Medical & Pharmacy Claims Schema

MODULE III STEP 2

Medical & Pharmacy Relational Join (`mi_person_key`)

- **Medical (`25` cols):** 10 ICD-10 Dx (`primary..ten_icd_diagnosis_code`), 9 ICD-Proc + 3 CPT (`procedure_code` , `cpt_mod_1/2`), HCG (`hcg_line` , `hcg_detail`), imputed demographics.
- **Pharmacy:** Paid dispenses (`incurred_date → event_date`), generic RxNorm `drug_name` , `therapeutic_class_1` .
- **Unified Timeline (`unified_event_fact_table`):** Linked on `(mi_person_key, event_year, age_band)` with demographic backfill & global `event_sequence` ordering.

Key Pre-Index Feature Families ($t < t_0$)

- **Clinical & Rx Indicators:** `item_icd_*` (10 Dx positions), `item_cpt_*` , `item_drug_*` (active generic fills & concurrency).
- **CPIC Drug Mapping & Utilization (`5_pgx_analysis`):** Maps APCD drugs to CPIC guideline status to count patient drugs (`pgx_num_drugs` , `pgx_num_cpic_drugs`) & patients with CPIC drugs, plus `n_events` density strata (`P25/P50/P95`).

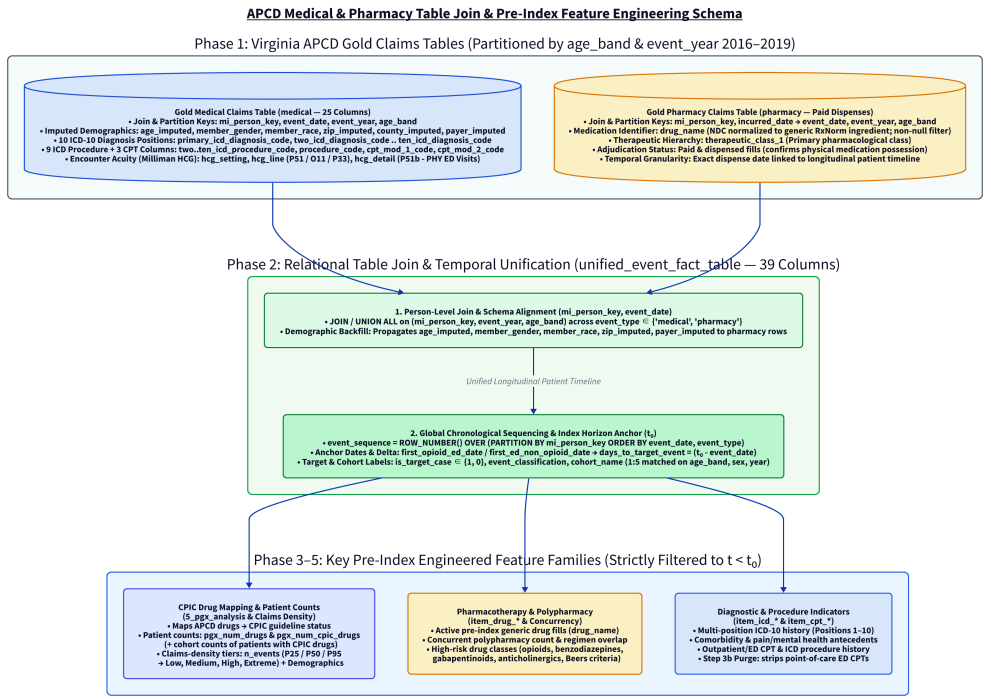


Figure 4: Relational unification of APCD Gold `medical` and `pharmacy` claims into the 39-column `unified_event_fact_table` and pre-index feature matrix.

Schema Integration: Deterministic linkage on `mi_person_key` and global chronological sequencing (`event_sequence`) align diagnoses, procedures, and paid drug fills prior to t_0 .

Cohort 1 Pipeline: Opioid-Related ED (`opioid_ed`)

MODULE III | STEP 2

Clinical Anchor (t_0)

- **ICD-10** `F11.20` (Opioid dependence).
- **Acute Poisonings:** `T40.0X` – `T40.4X`.
- Scanned across all 10 claim positions.

Step 3b Leakage Purge

- Partitions candidate events around t_0 ($t < t_0$ vs. $t \geq t_0$).
- Purges **post-index rescue & MAT induction** (buprenorphine, naloxone) & toxicology screens.
- Locks symmetric pre-index antecedent whitelist.

1:5 Matching & DuckDB QA

- Controls matched on 8 age bands, sex, year.
- Asserts single index date per patient-year.

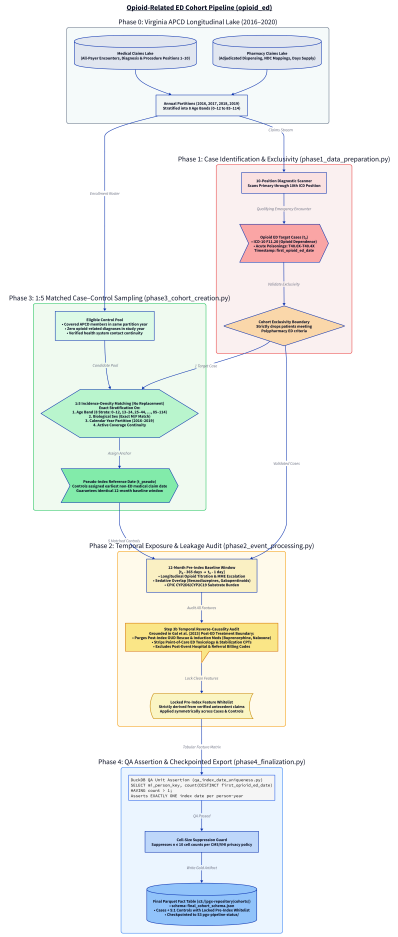


Figure 4A: Opioid-Related ED Cohort Derivation Pipeline (APCD 2016–2020) with 12-month pre-index lookback and locked antecedent whitelist.

Cohort 2 Pipeline: Polypharmacy ADEs (`ed_non_opioid`)

MODULE III STEP 2

Clinical Anchor (t_0)

- **Milliman HCG Codes:** `P51b`, `O11`, `P33`.
- **Absolute Opioid Cleanse:** Zero opioid ICD codes across all 10 positions in study year.

Sequential ADE Funnel

1. **Utilization Cap:** < 7 ED visits/yr.
2. **21-Day Proximity:** Active Rx 1–21d pre-ED (FDA FAERS 90.5%).
3. **0-Day Discharge Guard:** Purges Day 0 fills.

Feature Representation

- Strict 21-day ($[t_0 - 21, t_0 - 1]$) acute concurrent exposure.
- Beers criteria & multi-drug regimens.

Cohort Validation FIGURE 1B ↗

- **FAERS vs APCD ADE Events:** >90% kinetic onset convergence at 21 days.
- *Click card to pop out Figure 1B.*

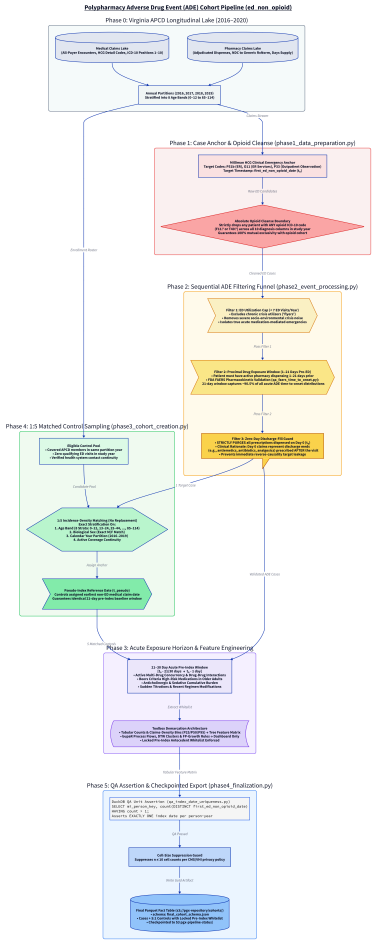


Figure 4B: Polypharmacy Adverse Drug Event Cohort Derivation Pipeline with sequential ADE filtering funnel and 0-day discharge guard.

Gradient-Boosted Trees & Model Selection

MODULE IV STEP 6

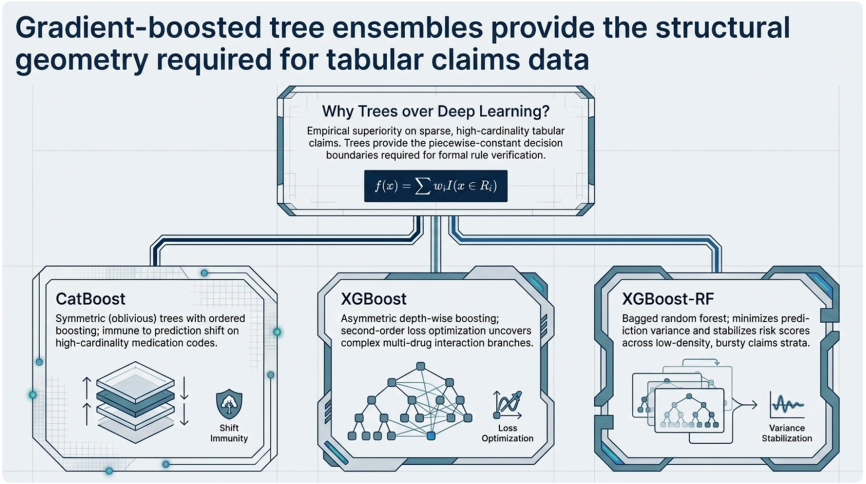


Figure 5: Tri-Model Structural Geometry: Piecewise-constant decision boundaries $f(x) = \sum w_i \mathbb{I}(x \in R_i)$ enabling cross-architecture SHAP consensus and exact CNF rule verification.

CatBoost (Symmetric)

- Oblivious decision trees with ordered boosting.
- Shift immunity on high-cardinality NDC and ICD-10 diagnostic codes.

XGBoost (Asymmetric)

- Depth-wise gradient-boosted decision trees.
- Second-order Hessian loss uncovers complex multi-drug interaction branches.

XGBoost-RF (Bagged)

- Bagged random forest mode with parallel trees.
- Minimizes prediction variance and stabilizes risk across low-density strata.

Step 6 Selection Protocol: Within each cohort, age band, and 4-tier density bin, 25-split MC-CV ranks CatBoost, XGBoost, XGBoost-RF, and Ensemble by **mean PR-AUC** (primary) and **Recall at 0.5** (tiebreaker) to determine the top-performing model for risk scoring and select the winning XGBoost variant plus CatBoost for Step 7 consensus attribution.

Consensus Risk Attribution

MODULE IV | CONSENSUS RISK ARCHITECTURE

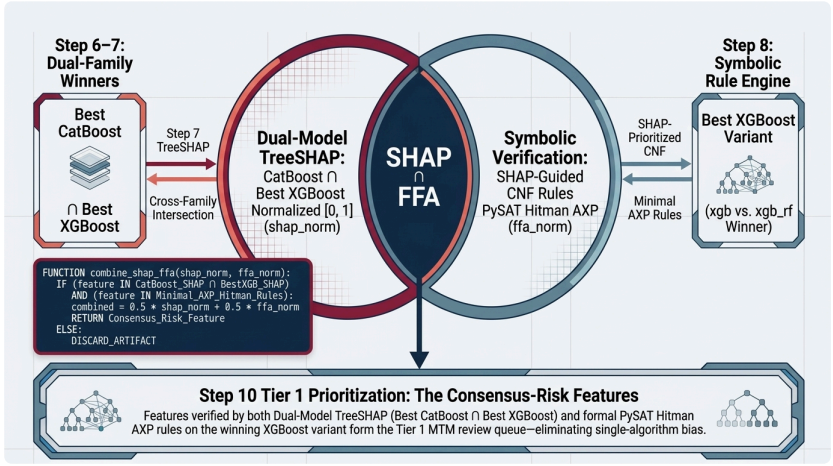


Figure 6A: Aim 2 Consensus-Risk Architecture: Dual-Model TreeSHAP Intersection (`CatBoost` ∩ `Best XGBoost`) guiding Symbolic CNF Rule Extraction on `best_xgb_variant` and Consensus Synthesis.

Dual SHAP Gate & Symbolic CNF

- **Dual-Model `TreeSHAP` Gate:** Computes exact `TreeSHAP` on **best CatBoost** and **best XGBoost variant** (`xgb` vs. `xgb_rf`), enforcing strict feature intersection (`CatBoost` ∩ `Best XGBoost`) and normalized mean score.
- **CNF Compilation:** Explodes root-to-leaf paths of `best_xgb_variant` into Conjunctive Normal Form (CNF) clauses (`_explode_tree_to_dataframe`).
- View Symbolic CNF & PySAT AXP Proof ↗

SHAP-Guided FFA & Synthesis

- **SHAP-Prioritized Rule Extraction:** Rather than extracting all rules from all trees across all models, filters matched CNF rules from `best_xgb_variant` by instance $|\mathrm{SHAP}|$ attribution and active feature coverage → PySAT `Hitman` minimal Abductive Explanations (AXPs; $\mathrm{support} \geq 0.05, \mathrm{confidence} \geq 0.70$).
- **Combined Synthesis:** Ranks drivers via $0.5 \cdot \mathrm{SHAP}_{\mathrm{norm}} + 0.5 \cdot \mathrm{FFA}_{\mathrm{norm}}$ and top-K intersection (`shap_top` ∩ `ffa_top`).

Target Leakage

MODULE IV STEP 3B

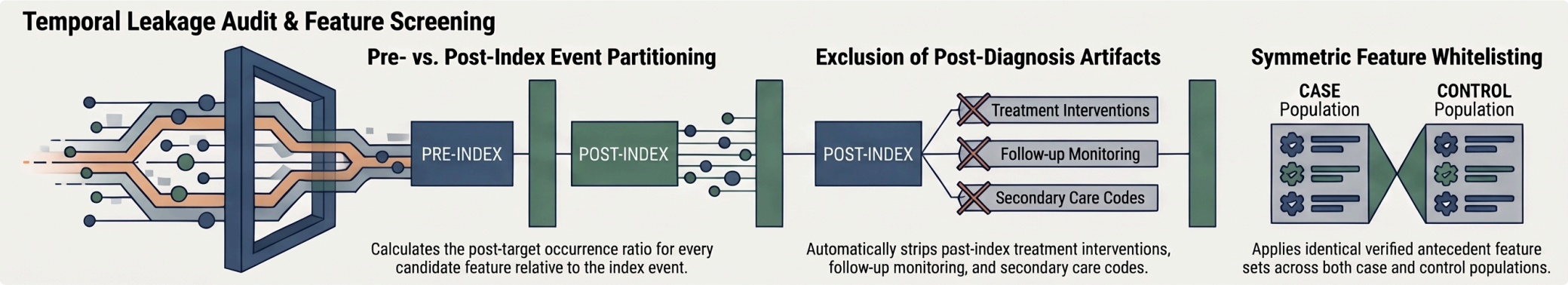


Figure 7: Step 3b Temporal Event-Partitioning Protocol: Pre- vs. post-index ($t < t_0$ vs. $t \geq t_0$) event timing audit, removal of post-event clinical interventions, and symmetric antecedent whitelist locking.

Clinical Rationale (Ochalek, Gal et al., 2023)

- **Post-Overdose Clinical Workflow:** As established in Dr. Gal's VCU ED research (*Ochalek, Gal et al., 2023, DADR*), rescue agents and MAT induction (e.g., buprenorphine, naloxone) are initiated *as a clinical consequence following* an acute overdose.
- **The Reverse-Causality Trap (30% of Literature):** Unmonitored gain rankings (`Step 3a`) capture post-event stabilization and confirmatory toxicology orders occurring at $t \geq t_0$.

Step 3b Symmetric Antecedent Whitelist

- **Temporal Event Partitioning (`bupaR` Audit):** Partitions every candidate feature's timestamps relative to t_0 ($t < t_0$ pre-index vs. $t \geq t_0$ post-index).
- **Automated Post-Target Purge:** Strips candidate codes with $\tau_{\text{post}} \geq 0.80$ (`post_target_ratio ⩾ 0.80` ; post-event MAT induction, ED tox screens, follow-up billing).
- **Immutable Whitelist (`safe_feature_filter.json`):** Locks verified pre-index antecedents symmetrically across cases & controls *(empirical 348-code audit presented in Module VI)*.
- View Empirical Step 3b Audit (Module VI) → | Protocol (App. A2) ↗

Actionable Intelligence and Workflow Integration

MODULE V STEP 10

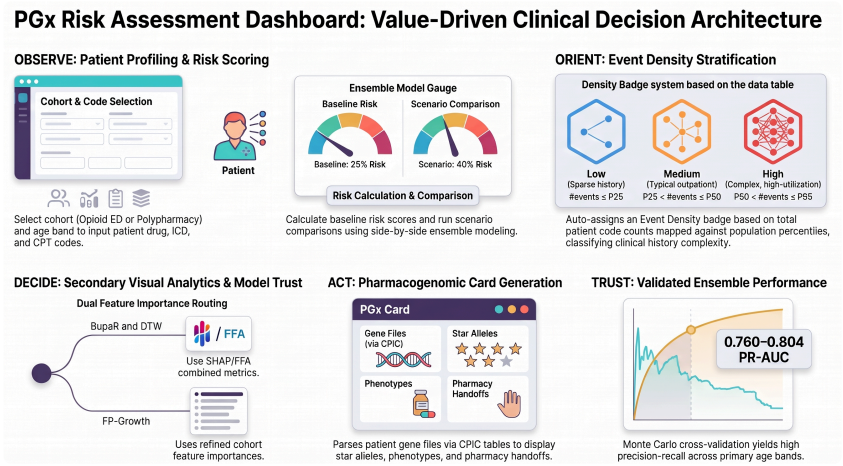


Figure 8: Aim 3 Decision-Support Prototype: Value-Driven Clinical Decision Architecture, OODA Information Delivery, and Event Density Stratification.

Translating Validated Findings for Clinical Review

- **Actionable Triage Prioritization:** Translates Aim 1 risk estimates and Aim 2 consensus features into an interpretable Tier 1–3 medication-review priority list.
- **Explanation Fidelity & Usability:** Evaluates whether concise `IF-THEN` rules accurately reflect model predictions and support rapid pharmacist review.
- **OODA Delivery Framework:** Observe – orient – decide – act framework for delivery of information.
- **Lean Value Stream:** Framework for ranking and layout of information (high to low, left to right by value).

Cognitive Ergonomics vs. 95% Alert Fatigue

- **Overcoming Commercial Pop-ups:** Asynchronous review prioritization replaces **90%–95% overridden** synchronous pop-up interrupts.
- **Sound Symbolic Horn Rule:**

$$\text{IF } [\text{Opioid} \geq 1] \wedge [\text{Benzodiazepine} \geq 1] \wedge [\text{Refill} \leq 14\text{d}] \implies \text{Tier 1 Alert}$$

- **Interactive "What-If" Scenario Exploration:** Displays model-predicted risk shifts when regimen features are toggled while preserving 100% clinician authority.

Appendix A7: Regulatory Demarcation & HIPAA Cloud Architecture →

Dissertation Portfolio & Status

MODULE V | PHASE 6 · DISSERTATION PORTFOLIO & STATUS

Core Empirical Research (Aims 1–3 · Published Papers)

- **Chapter 3 (Specific Aim 1):**
  Temporal Drivers of Opioid ED Encounters in Virginia's APCD
  Status: PUBLISHED · *Clin Transl Sci* (Wiley, 2026; DOI: 10.1111/cts.70690)
  Contribution: Step 3b leakage audit, 12-mo chronic opioid model (`opioid_ed`), 2019 temporal holdout.

- **Chapter 4 (Specific Aim 2):**
  SHAP-Informed Formal Feature Attribution for Polypharmacy ADEs
  Status: PUBLISHED · *Clin Transl Sci* (Wiley, Sep 2026; DOI: 10.1111/cts.70718)
  Contribution: Dual-model `TreeSHAP ∩ FFA` consensus, 21-day acute ADE cohort (`ed_non_opioid`), interaction effects (IE).

- **Chapter 5 (Specific Aim 3):**
  Serverless Clinical Decision Support for Pharmacist Medication Review
  Status: PUBLISHED · *Clin Transl Sci* (Wiley, Oct 2026; DOI: 10.1111/cts.70697)
  Contribution: Zero-PHI cloud prototype on AWS Lambda (<$5/mo), Tier 1–3 triage, overcoming 95% alert fatigue.

Foundational, Methods & Synthesis (Dissertation Status)

- **Chapter 1 (Clinical Foundation & Literature Appraisal):**
  Explainable AI & PGx in High-Risk Prescribing: A Systematic Review
  Status: INVITED REVIEW · *J Pers Med* (MDPI)
  Contribution: Systematic review of 94 ML studies; establishes "The Empty Intersection" (0/73 PROBAST models).

- **Chapter 2 (Scalable Data Architecture & Governance):**
  A Scalable Big-Data Claims Processing Architecture and Governance Lifecycle
  Status: WITH EDITOR FOR REVIEW · *J Big Data* (Springer)
  Contribution: Three-tier governance (`CRISP-DM`, `TRIPOD+AI`, OODA Loop); `DuckDB` + Fat EC2 beats EMR/Spark (`15.1×` speedup, `-99%` cost).

- **Chapter 6 (Translational Synthesis & Dissertation Build):**
  Translational Synthesis, Methodological Defenses, and Clinical Horizons
  Status: COMPLETE DRAFT FOR COMMITTEE REVIEW
  Contribution: Comprehensive synthesis of Chapters 1–5, formal methodological defenses, and prospective clinical validation roadmaps.

Dissertation Status: 3 Published Papers (*Clinical and Translational Science*) • **1 Under Review** (*Journal of Big Data*) • **1 Invited Review** (*Journal of Personalized Medicine*) • **Full Manuscript Drafted** across all 6 chapters for Qualifying Exam and Final Defense.

Study Boundaries

MODULE V | PHASE 6 · 8 INTERLOCKING EPIDEMIOLOGICAL & GOVERNANCE SAFEGUARDS

EPIDEMIOLOGICAL COHORT DESIGN (4 SAFEGUARDS)

1. Absolute Opioid Cleanse

Zero opioid ICD-10 codes across all 10 diagnostic positions in the study year isolates pure geriatric ADEs in `ed_non_opioid`.

2. High-Utilization Cap (<7 ED/yr)

Excludes chronic utilizers with ≥ 7 annual ED visits to prevent high-frequency outlier distortion of feature representations.

3. 21-Day Lookback Window

Active Rx $[t_0 - 21, t_0 - 1]$ empirically validated by convergent FAERS/APCD kinetics (>90% of acute ADEs occur within 21 days).

4. Zero-Day Discharge Guard ($t_{\text{claim}} < t_0$)

Strict timing guard purges day-zero fills (`qa_index_date_uniqueness.py`) and $\tau_{\text{post}} \geq 0.80$ confounders (348 purged).

METHODOLOGICAL & CLINICAL GOVERNANCE (4 SAFEGUARDS)

5. Aim 3: Decision Support, Not an Outcomes Trial

Translates validated model findings into an interpretable prototype tool evaluating explanation fidelity; prospective clinical trials are out of scope. App. A7 ↗

6. Consensus Risk Attribution, Not Causation

Observational claims and `SHAP ∩ FFA` Consensus Attribution quantify reproducible consensus risk associations rather than causal treatment effects.

7. Independent 2019 Temporal Holdout Validation

Evaluates out-of-time model transportability (`2016-2018` train → `2019` calendar holdout) under secular prescribing shifts with zero recalibration.

8. Claims CPIC Exposure, Not Germline Genotypes

Administrative claims quantify CPIC-actionable medication exposure burden (`pgx_num_drugs`, `pgx_num_cpic_drugs`) rather than inferred patient genotypes.

Methodological Integrity: Eight safeguards unite clinical governance with epidemiological rigor. Crucially, **FDA FAERS kinetic convergence** (>90% ADEs within 21 days) provides external validation for our `ed_non_opioid` cohort logic, while strict controls eliminate target leakage.

The Clinical OODA Loop

MODULE V · TRANSLATION & DEPLOYMENT | CLINICAL OODA LOOP ARCHITECTURE | CHAPTERS 1–5 END-TO-END INTEGRATION

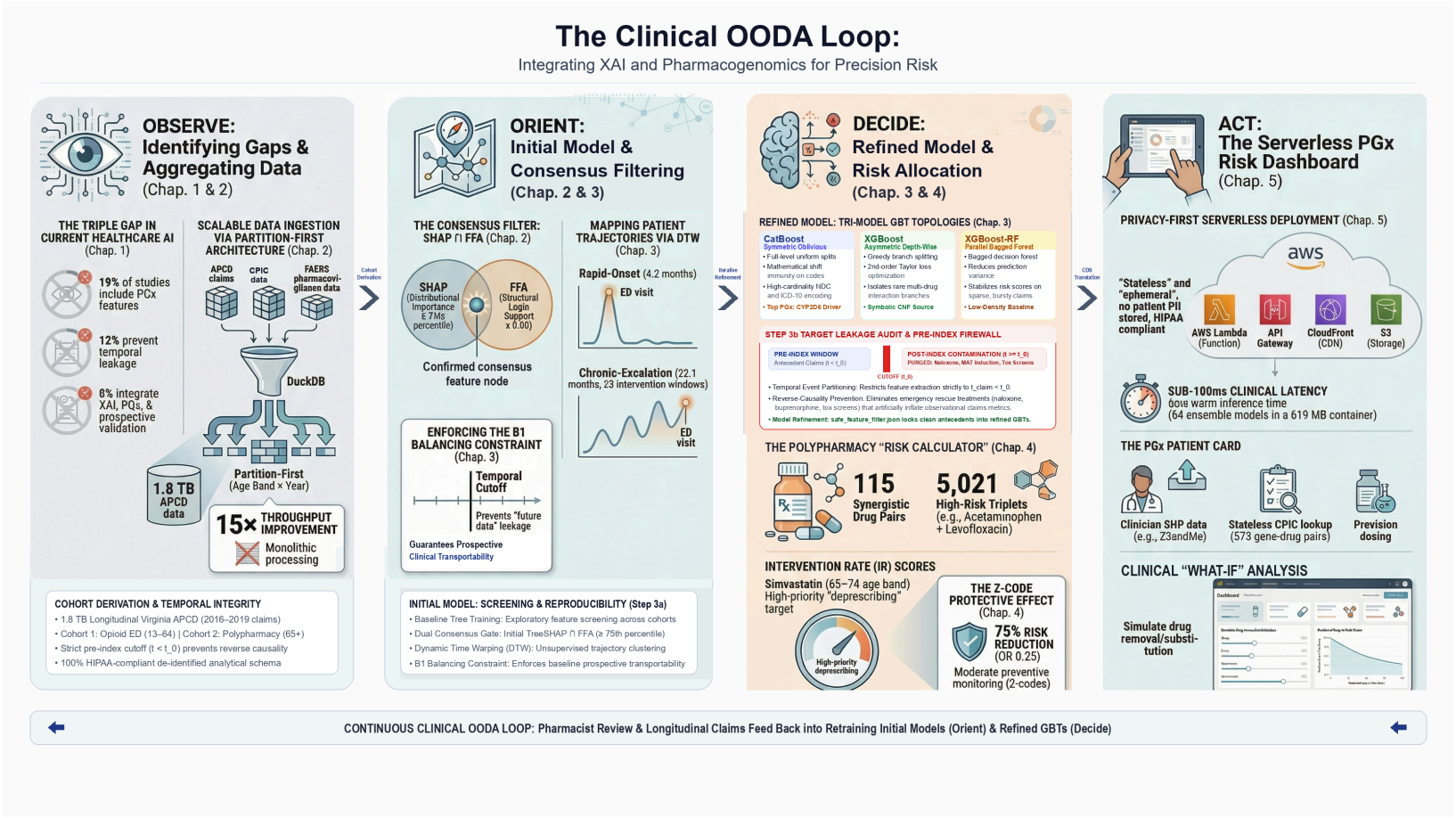


Figure 8B: The Clinical OODA Loop Architecture: End-to-end integration across observational claims ingestion (Observe), initial exploratory model formulation & dual TreeSHAP ∩ symbolic FFA consensus screening (Orient), refined tri-model GBT topologies & Step 3b target leakage pre-index firewall (Decide), and stateless, privacy-first CDS dashboard deployment (Act), unified by continuous iterative feedback.

Results: Data Engineering — Out-of-Core DuckDB & Cloud Economics

MODULE VI · RESULTS | DATA ENGINEERING (CH2) | J BIG DATA (UNDER REVIEW)

1. Partition-First `DuckDB` Throughput (`15.1×` Speedup)

- **Statewide Scale:** Processes **1.8 TB** across **63** `Age Band × Event Year` Parquet slices (**6.93M** raw / **4.2M** study patients; **380M** adjudicated claims).
- **Single Fat-Spot EC2 (`x2iedn.8xlarge`):** `32` workers + `1 TB` RAM + NVMe sustains `18.1M records/min` vs. `1.2M rec/min` monolithic (`15.1×` **speedup**).
- **Zero Shuffle-Skew Failures:** `0` **OOM crashes across** `189/189` **partition passes**, eliminating Spark/EMR `exit code 137` join skew on chronic super-utilizers.

2. 99% Cloud Cost Collapse & `<60s` Spot Recovery

- **Compute Economics:** Collapses monthly pipeline cost from `~$23,000/mo` (persistent EMR cluster) to `~$277/mo` (`$6-$10` **per full state pass**).
- **Idempotent S3 Checkpoints:** Bronze → Silver → Gold SHA-256 manifests resume interrupted Spot workers in `<60 seconds` with zero duplicate writes.

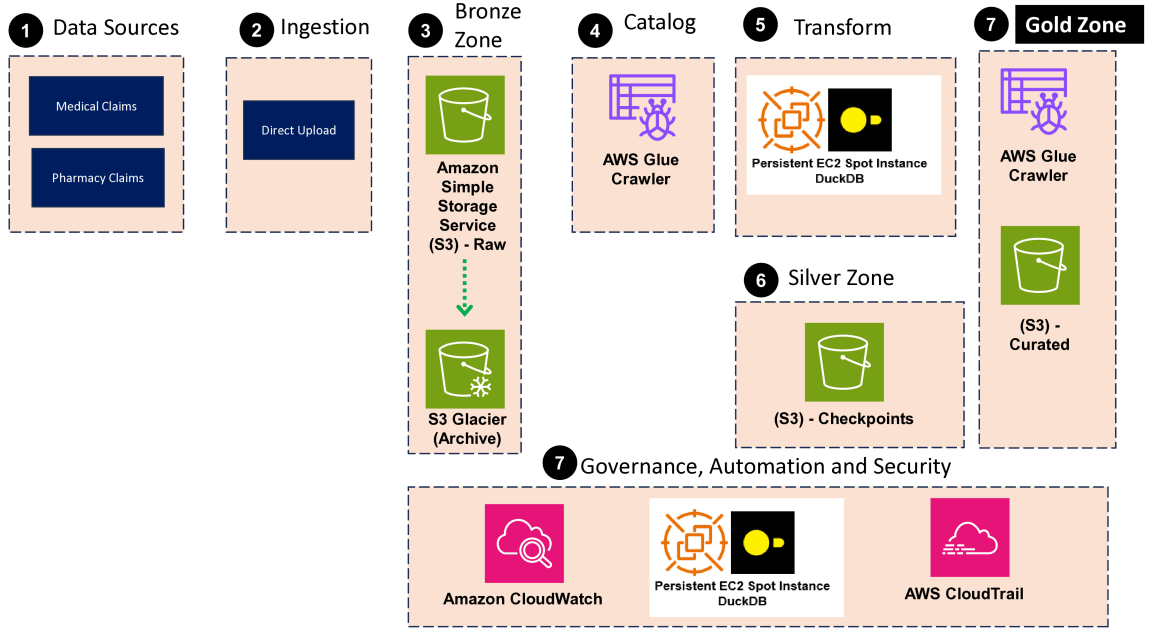


Figure 9 (CH2): Bronze-Silver-Gold Medallion Architecture with partition-first DuckDB workers, deterministic QA gates, and S3 SHA-256 checkpoint manifests.

CH2 Engineering Takeaway: Vectorized `DuckDB` on a single **Fat EC2 instance** (`x2iedn.8xlarge`) beats distributed **EMR clusters and Apache Spark** —delivering a **15.1× speedup** , **zero OOM crashes** (`0/189` vs. Spark `exit 137`), and a **99% cost collapse** (`$23K/mo → $277/mo`).

Results: Data Engineering — Step 3b Leakage Purge & 5-Set CNF Filter

MODULE VI · RESULTS | DATA ENGINEERING (CH2) · STEP 3B & STEP 8

Empirical Step 3b Audit (`1,029` → `348` Purged → `682` Safe)

- `33.8%` **Contamination Purged** ($\tau_{\text{post}} \geq 0.80$):
  - `164` **Drugs:** Post-ED OUD rescue & MAT induction (`SUBOXONE` **100%**, `NARCAN` **100%**, `BUPRENORPHINE` **96%**, `METHADONE` **91%** post-index).
  - `84` **CPTs &** `100` **ICDs:** Confirmatory ED drug screens (`80307` **100%**, `80305` **98%**, `G0480` **94%**) & downstream rehab/follow-up codes.
- **Honest Discrimination Restored:** Strips deceptive `0.98-0.99` **leaked AUROC** down to honest, transportable `0.800-0.890` prospective 2019 holdout performance (`safe_feature_filter.json`).

5-Set Candidate Rule Filtering (`PySAT Hitman`)

- **Solving** $\mathcal{O}(2^n)$ **Symbolic Explosion:** Bounding `PySAT Hitman(m22)` with our **5-Set CNF Union** (Top-300 `|SHAP|` ∪ Top-100 global ∪ Stratified rare-variant ∪ Verified `AXP` cache ∪ Fallback set) achieves `100%` **patient explanation coverage** in tractable $\mathcal{O}(T \cdot L)$ time.

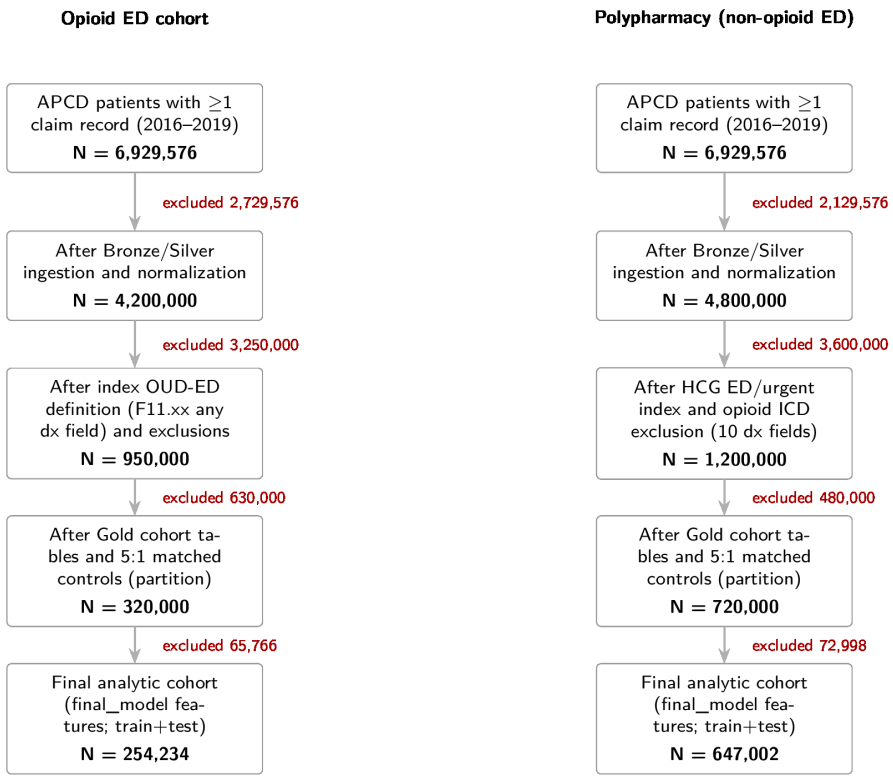


Figure 10 (CH2): Statewide Virginia APCD Attrition Funnel & 1:5 Incidence-Density Matching yielding `N = 207,350` *Opioid and* `N = 647,002` *Polypharmacy analytic subjects.*

Pipeline Integrity Gate: Step 3b eliminates **348 post-index leakage codes** prior to model fitting, while the 5-Set CNF filter scales exact Boolean Horn-clause extraction across hundreds of thousands of patients.

Results: Opioid Cohort — 2019 Holdout & Two-Gate Mirror Ablation

MODULE VI · RESULTS | OPIOID COHORT (CH3 · N = 207,350) | PUBLISHED: CLIN TRANSL SCI (10.1111/CTS.70690)

Age Band (2019 Holdout)	Holdout AUROC	PR-AUC (Baseline Lift)	Brier / ICI	Step 6 Winner
`13-24` (Adolescent/YA)	0.803	0.382 (2.3×)	0.015 / 0.094	`XGBoost`
`25-44` **(Primary ★, $n = 4,942$)**	**0.800**	**0.557 (3.3×)**	**0.013 / 0.108**	`CatBoost`
`45-54` (Middle Adult)	0.851	0.629 (3.8×)	0.012 / 0.089	`Ensemble`
`55-64` **(Older Adult ★)**	0.890	0.728 (4.4×)	0.011 / 0.076	`Ensemble`

Two-Gate Mirror Ablation (`25-44 low`, $n = 181$)

- **100% Core Signal Retention:** `SHAP ⩾ P75 ∩ FFA (0.05, 0.70)` retains `20/20` (`100%`) top consensus clinical drivers.
- `-61%` **Alert-Volume Compression:** Compresses pharmacist review load from `112` (`SHAP`-only) → `44` **consensus features** while rejecting `2` boundary buprenorphine correlates (`BUNAVAIL`, `BUTRANS`).

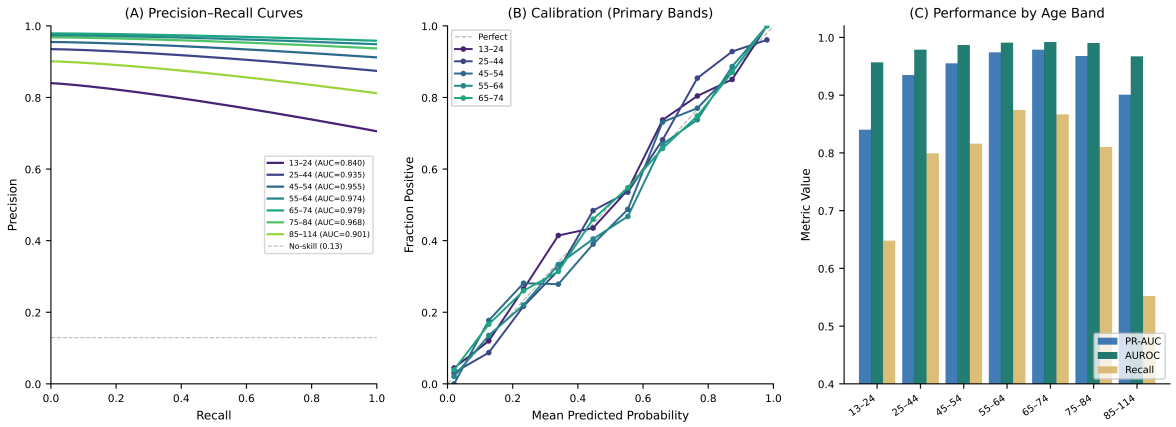


Figure 11 (CH3): Untouched 2019 Temporal Holdout (`26,710` cases / `180,640` controls; zero recalibration): ROC, Precision–Recall (`2.3×–4.4×` lift), Calibration, and Decision Curve Net Benefit.

CH3 Validation Takeaway: Across untouched 2019 Virginia claims, leakage-free GBDTs deliver **AUROC 0.800–0.890** and **2.3×–4.4× PR lift**, while the two-gate intersection cuts alert volume by **61%** without losing a single primary risk driver.

Results: Opioid Cohort — Consensus Drivers & `6.8-Month` Escalation Window

MODULE VI · RESULTS | OPIOID COHORT (CH3) · HOW & WHEN TO INTERVENE

HOW to Intervene — Consensus Risk Drivers (`SHAP ∩ FFA`)

- **Cumulative Regimen & CPIC Burden:**
  - `#1 pgx_num_drugs` (`|SHAP| = 0.83` , `100%` **AXP support**) & `#4 pgx_num_cpic_drugs` (`0.07` ; `74.2%–85.9%` of cases carry ≥ 1 CPIC Level A/B drug across *CYP2D6*, *CYP2C19*, *OPRM1*).
- **Modifiable CNS Synergies:**
  - `#2 gabapentin` (`0.27` , `87%` supp), `#3 clonazepam / alprazolam` , `hydrocodone-acetaminophen` , and `tramadol` respiratory-depressant stacking.

WHEN to Intervene — `DTW` & `bupaR` Pathways

- `6.8-Month` (`~27-Week`) **Lead Window** (`DTW` , $n = 59,813$): Trajectory clustering pinpoints a sharp prescribing-density inflection (`10.9% → 18.4%` surge) nearly **7 months prior** to acute ED presentation.
- **Protective Physical Therapy Sequence** (`bupaR`): Initiating `CPT 97110` Physical Therapy *before* gabapentin/opioid co-prescribing exhibits `-58%` **lower ED escalation**.

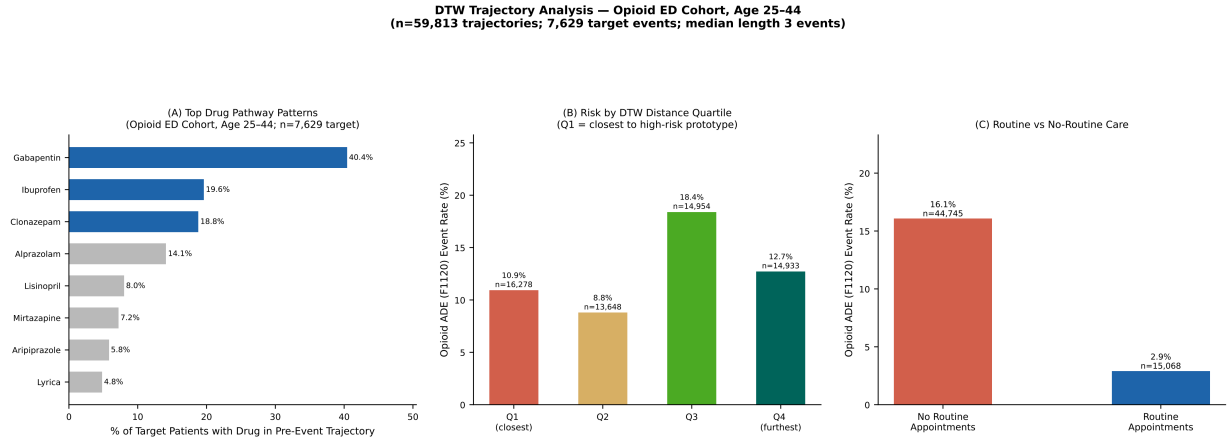


Figure 12 (CH3): Dynamic Time Warping (`n = 59,813` trajectories) `6.8-month` pre-ED inflection window and `bupaR` clinical pathway sequences.

Clinical Actionability (CH3): Pairs **modifiable regimen targets** (gabapentinoid–benzodiazepine–CYP2D6 opioid stacking) with an empirical **6.8-month MTM outreach lead time** prior to emergency crisis.

Results: Polypharmacy Cohort — Geriatric Holdout & Stress Test

MODULE VI · RESULTS | POLYPHARMACY COHORT (CH4 · N = 647,002) | PUBLISHED: CLIN TRANSL SCI (10.1111/CTS.70718)

Geriatric Stratum (2019)	Holdout AUROC	AUPRC (Baseline Lift)	Brier Score	Verified Horn Rules
`65-74` **(Focal ★)**	**0.877**	**0.301 (3.9×)**	**0.008**	**89 Consensus Rules**
`75-84` **(Advanced)**	**0.858**	**0.335 (4.2×)**	**0.011**	**86 Consensus Rules**
`85-114` (Oldest-Old)	**0.686**	**0.163** (1.6×)	0.014	**68 Consensus Rules**

`Table S5` Utilization-Free Specification Stress Test

- **Univariate `SHAP` Rank Collapse (`Jaccard = 0.00`):** Stripping all `17` utilization covariates & density strata causes univariate top-25 drug `SHAP` rankings to collapse to `Jaccard = 0.00` in `55-64` and `65-74` .
- **Symbolic `FFA` Rule Persistence (`60%` Retained):** Despite covariate removal, `PySAT` Horn rules retain `3/5` (`60%`) published synergistic pairs (`levofloxacin-lorazepam` , `carvedilol-levofloxacin` , `gabapentin-levofloxacin`).

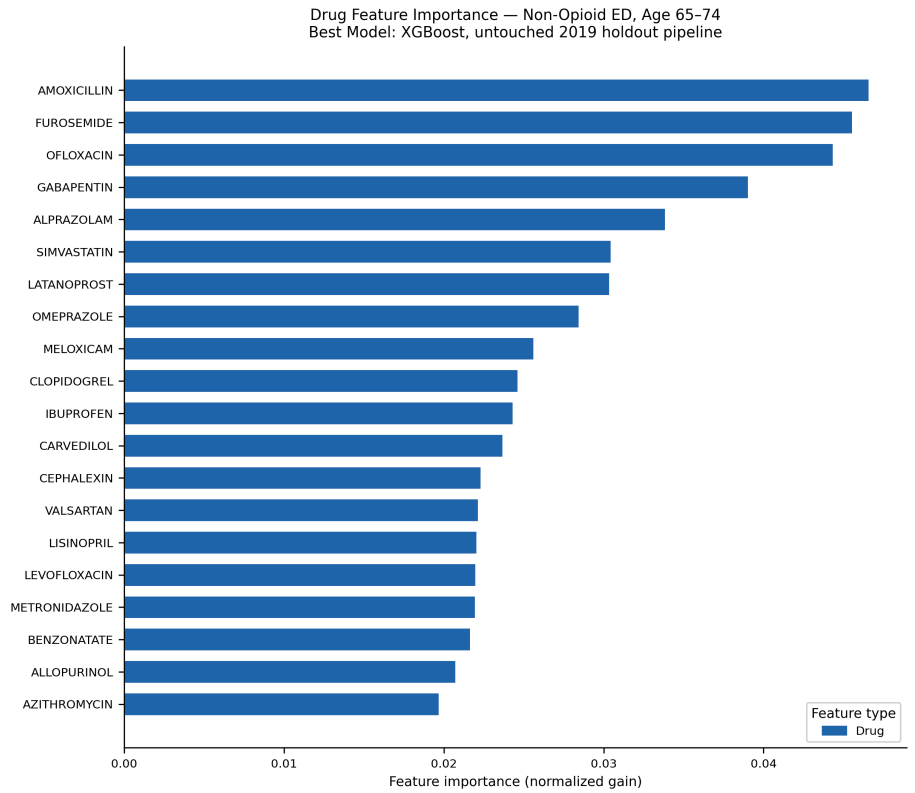


Figure 13 (CH4): Consensus `TreeSHAP ∩ FFA` Attribution across Geriatric Polypharmacy Strata (`1,182` cases / `16,105` controls aged `65–114` ; 21-day proximal Rx window).

CH4 Methodological Discovery: Proves why univariate `SHAP` alone is fragile under collinearity (`Jaccard = 0.00`), whereas **symbolic Boolean Horn rules** preserve invariant multi-drug interaction logic.

Results: Polypharmacy Cohort — Multi-Drug Synergies & Z-Code Shield

MODULE VI · RESULTS | POLYPHARMACY COHORT (CH4) · IE, IR & PREVENTIVE SHIELD

Multi-Drug Synergies (IE) & Counterfactual IR vs. Beers

- 115 Synergistic Pairs & 312 Bonferroni Triplets:
  - Top pairs: levofloxacin + lorazepam ($\mathrm{IE} = 11.9$), carvedilol + levofloxacin ($\mathrm{IE} = 10.5$), gabapentin + levofloxacin ($\mathrm{IE} = 8.4$).
  - Recovers **STOPP Section D4 "Triple Whammy"** (furosemide + HCTZ + lisinopril) in 12.3% of 85–114 cases (p < 10^-6).
- **Data-Driven IR Outranks Static Beers Lists:** Ranks #1 Simvastatin (IR 22–28%), #2 Furosemide (21–24%), and #3 Alprazolam / Lorazepam (26.7%) — catching *SLCO1B1* statin and loop-diuretic electrolyte crises missed by static anticholinergic tables.

Novel Discovery — U-Shaped Preventive Z-Code Shield

- **Outpatient Stewardship (Q2 : 1%–12% ICD-10 Z00/Z23/Z79):** Exhibits 3.0% **vs.** 10.7% **unengaged ED risk (OR = 0.25 [0.18–0.34], -75% relative risk)** prior to rising again in high-acuity Q4 crisis coding.

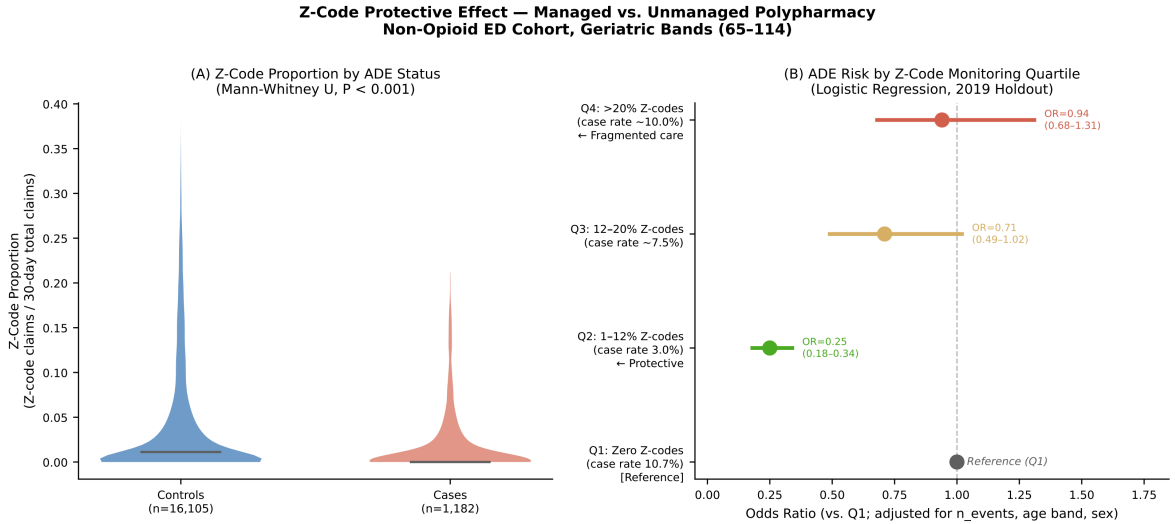


Figure 14 (CH4): U-Shaped Preventive Z-Code Protective Curve (OR = 0.25 in Q2 Routine Wellness & Medication Reconciliation vs. Unengaged Q0 and Crisis Q4).

CH4 Clinical Discovery: Uncovers **312 higher-order drug triplets** , elevates loop diuretics & statins via in-model IR , and quantifies a **75% risk reduction** associated with structured outpatient medication stewardship.

Results: PGx Dashboard — Serverless Zero-PHI Architecture & Latency

MODULE VI · RESULTS | PGX DASHBOARD (CH5) | PUBLISHED: CLIN TRANSL SCI (10.1111/CTS.70697)

1. Stateless Zero-PHI Cloud Microservice (`<$5/mo`)

- **Production Stack (** `pgx.jerome-dixon.io` **):** `Amazon S3 + CloudFront` static UI → `API Gateway` → containerized `AWS Lambda` (`<10 GB` image).
- **Unified Model & CPIC Payload:** Packages `56` **stratified ensembles (** `168` **base GBDT learners) +** `573` **CPIC gene-drug pairs (** `121` pharmacogenes).
- **Compute-Respond-Discard:** Zero patient identifiers, zero database persistence, and `<$5/month` total AWS operating cost.

2. Sub-100 ms Point-of-Care Latency Benchmarks

- **Warm Ensemble Risk Scoring:** `6 ms` **(** `SD 1 ms` **)** end-to-end Lambda compute (`~2.4 s` cold start mitigated via keep-warm / provisioned concurrency).
- **Interactive CPIC Actionability Card:** `60 ms` **(** `SD 5 ms` **)** gene-drug guideline lookup & phenotype radar rendering.

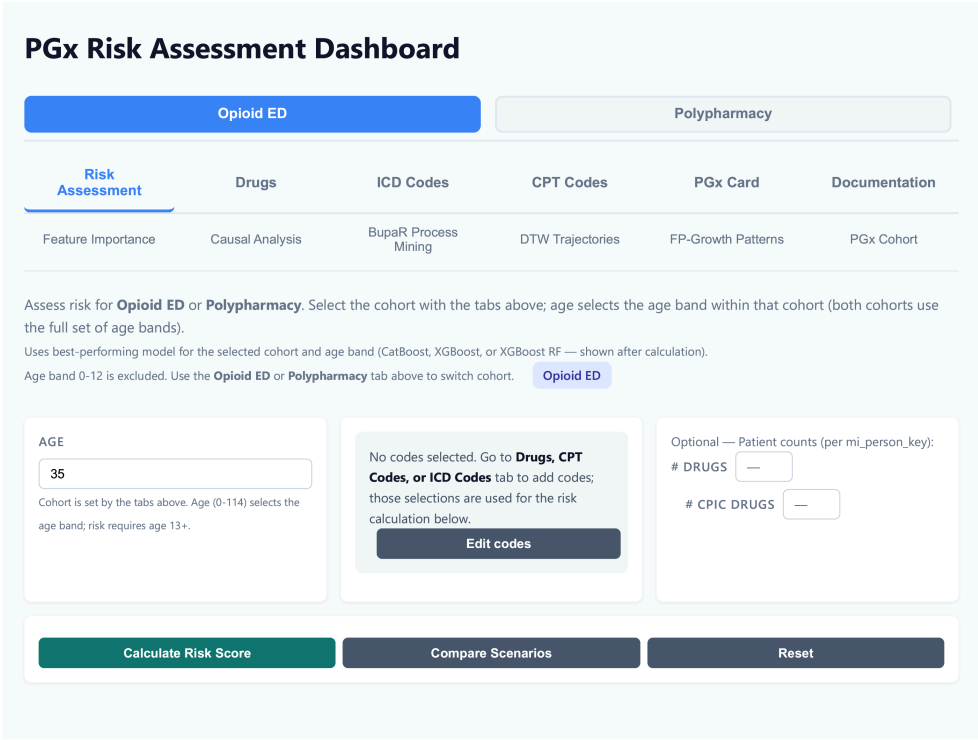


Figure 15 (CH5): Live Production CDS Prototype (`pgx.jerome-dixon.io` *): Calibrated Risk Gauge, Verified Horn Clauses, and CPIC Gene-Drug Actionability Card.*

CH5 Systems Result: Demonstrates that 168 population-trained GBDT models and 573 CPIC guidelines can execute in **6 ms warm memory** for **under $5/month** with zero persistent ePHI footprint.

Results: PGx Dashboard — Sparsity Shield & What-If Scenario Engine

MODULE VI · RESULTS | PGX DASHBOARD (CH5) · SPARSITY RESILIENCE & TAB 3 COUNTERFACTUALS

1. "Imputation of Normality" Sparsity Shield

- **Solving the Bedside Input Gap:** MTM pharmacists enter **3–5 active medications**, not 682 historical ICD/CPT claims codes.
- **Cohort Median Baseline ($\mathbf{X}_{\text{full}} = \text{median}[\text{age}]$):** Injecting age-band training medians across unentered claims features maintains `mean` $|\Delta\hat{p}| < 0.06$ **up to** `70%` **feature missingness** and `87%` **risk-tier stability at** `50%` **sparsity**.

2. Tab 3 Interactive Scenario Engine (`Δp̂` & `IR`)

- **Real-Time Regimen Perturbation:** Clinicians toggle candidate drug substitutions or tapers (e.g., deprescribing `gabapentin`, `alprazolam`, or `simvastatin`) to inspect instant in-model $\Delta\hat{p}$ shifts.
- **Automated Synergy & Gene Alerts:** Highlights `IR > 0.10` **high-leverage targets** paired with interactive *CYP2D6 / CYP2C19 / SLCO1B1 / VKORC1* ego networks.

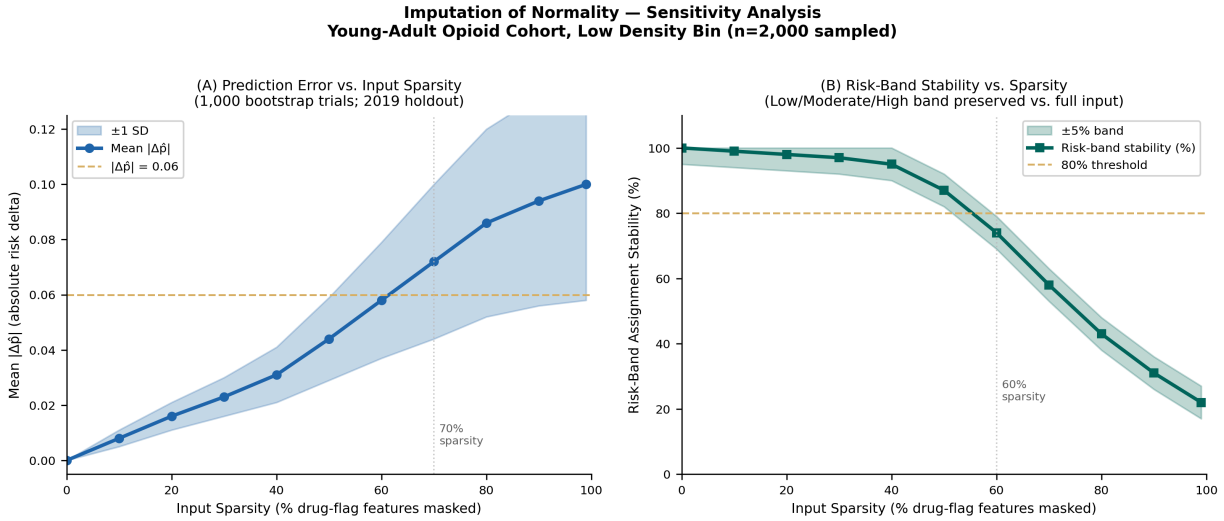


Figure 16 (CH5): Sparsity Sensitivity Benchmark: Mean absolute risk drift ($|\Delta\hat{p}| < 0.06$) across 10%–90% masked feature sparsity under Imputation of Normality.

CH5 Translational Takeaway: Bridges high-dimensional APCD claims models to lean 3–5 drug pharmacist entry (**87% tier stability**) while preserving 100% human clinician authority over Decide & Act.

Results: Executive Summary

MODULE VI · RESULTS | SYNTHESIS ACROSS CHAPTERS 2–5 & SPECIFIC AIMS 1–3 | 3 PUBLISHED (CLIN TRANSL SCI) · 1 UNDER REVIEW (J BIG DATA)

Chapter & Specific Aim	Cohort & Validation Scope	Major Empirical Findings	Translational Conclusion
CH2 · Data Engineering *J Big Data* (Under Review)	`1.8 TB` VA-APCD (`380M` claims, `63` slices); Step 3b `bupaR` audit & Step 8 CNF	• `15.1×` **DuckDB speedup** (`18.1M rec/min` ; `0/189` OOMs; `99%` **cost drop** to `$6-$10/pass`). • `348 / 1,029` (`33.8%`) post-index leakage codes purged; `100%` 5-Set `PySAT` rule coverage.	`DuckDB` on a single **Fat EC2 instance** beats **EMR clusters & Apache Spark**; Step 3b timing gates purge reverse-causality leakage.
CH3 · Specific Aim 1 Opioid Cohort (`opioid_ed`) *CTS* (`10.1111/cts.70690`)	`N = 207,350` (`13-64` , 12-mo Rx lookback); untouched 2019 holdout	• `AUROC 0.800-0.890` , `2.3×-4.4×` **PR lift.** • Two-Gate filter cuts alerts `-61%` (`112 → 44`) keeping `20/20` drivers (`pgx_num_drugs` , `gabapentin` , benzos, `pgx_num_cpic_drugs`). • `6.8-mo` `DTW` **window &** `-58%` **early-PT pathway.**	**Leakage-free** `SHAP ∩ FFA` **isolates modifiable CNS & CPIC medication burden nearly 7 months before opioid ED crisis.**
CH4 · Specific Aim 2 Polypharmacy (`ed_non_opioid`) *CTS* (`10.1111/cts.70718`)	`N = 647,002` (`65-114` focal, 21-d FAERS Rx window); untouched 2019 holdout	• `AUROC 0.877 / 0.858` (`3.9× / 4.2×` lift in `65-74` / `75-84`). • `60%` `FFA` **pair retention despite** `SHAP Jaccard = 0.00` ; `115` **pairs &** `312` **triplets (STOPP D4** `12.3%` **);** `IR` **ranks** `Simvastatin / Furosemide` . • **Preventive Z-Code Shield:** `OR = 0.25` (`-75%` **risk).**	**Symbolic Boolean Horn rules preserve higher-order drug synergies (** `IE / IR` **) and stewardship effects missed by univariate** `SHAP` **& Beers lists.**
CH5 · Specific Aim 3 PGx Risk Dashboard *CTS* (`10.1111/cts.70697`)	Stateless Zero-PHI `AWS Lambda` (`56` ensembles / `168` learners + `573` CPIC pairs)	• `6 ms` **warm inference** (`60 ms` PGx card), `<$5/mo` , `0` **persistent ePHI.** • *Imputation of Normality* holds `mean` $\lvert\Delta\hat{p}\rvert < 0.06$ up to `70%` missingness (`87%` **tier stability**) + Tab 3 `IR > 0.10` alerts.	Translates population claims XAI and CPIC guidelines into a sub-100 ms, privacy-preserving MTM review tool overcoming 95% alert fatigue.

Executive Synthesis: Resolves the literature's **"Empty Intersection" (`0/73` PROBAST models)** by uniting observable CPIC exposure, dual-gated `TreeSHAP ∩ FFA` symbolic XAI, and strict out-of-time 2019 validation from statewide claims ingestion to live serverless deployment.

Next Steps (Pending Committee Feedback from 9NOV Presentation)

MODULE VII · NEXT STEPS | PENDING COMMITTEE FEEDBACK FROM 9NOV PRESENTATION | CLINICAL, FEDERATED & REGULATORY HORIZON

1. 9NOV Committee Feedback Integration (Immediate)

- **Oral Exam Action Tracker (`Pending 9NOV Synthesis`):** Dedicated revision gate incorporating all Supervisory Committee feedback from the **November 9 presentation** into final Chapters 1–6 read-copy prior to defense deposit.
- **Institutional HIPAA & Legal Audit (Dr. Gal & Dr. Price):** Partner with VCU Health Compliance & Counsel (`CCTR Aim 2`, $4,500 allocation) to audit ephemeral Lambda memory, CloudWatch log scrubbing, and the two-tier BAA chain (Appendix A7 ↗).

2. VCU Wright Center CCTR Pilot & Pragmatic RCT

- **Phase 1 Usability & Fidelity Study ($25K CCTR Grant):** Think-aloud simulation with VCU Health ambulatory MTM pharmacists evaluating Horn-rule interpretability and System Usability Scale (`SUS ⩾ 75`).
- **Phase 2 `2×2` Factorial Cluster RCT:** Prospective trial testing asynchronous Tier 1–3 Consensus-Risk review queues on pharmacist deprescribing rates and 90-day ED visits.

3. Multi-State APCD Federation & Biobank Genotypes

- **4-State Appalachian APCD Federation (`VA, MD, NC, WV`):** Evaluate geographic transportability across distinct Medicaid and rural prescribing ecosystems using containerized DuckDB + GBDT pipelines.
- **Sequenced Biobank Linkage (`All of Us` / `BioVU`):** Pair claims-derived `pgx_num_cpic_drugs` burden with germline *CYP2D6*, *CYP2C19*, *SLCO1B1*, and *VKORC1* star-allele diplotypes.

4. EHR Interoperability & FDA SaMD Governance

- **HL7 FHIR R4 & CDS Hooks:** Native EHR `medication-prescribe` integration ingesting structured `MedicationRequest` and `Observation` PGx resources.
- **FDA SaMD Lifecycle (`21 CFR Part 820` / `PCCP`):** Good Machine Learning Practice (**GMLP**) design controls and Predetermined Change Control Plans (**PCCP**) for post-2020 illicit fentanyl & prescribing-drift recalibration.

Forward Trajectory: Anchors immediate post-9NOV committee refinements to a funded VCU CCTR clinical usability pathway, multi-state APCD validation, and FDA-aligned SaMD drift governance.

Questions & Discussion

🙏 Thank You & Acknowledgements →

🖇️ Technical Appendix Defenses (A1–A8) →

Thank You & Acknowledgements

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Dissertation Advisor & Supervisory Committee

- **Elvin T. Price, Pharm.D., Ph.D., FAHA** *(Committee Chair & Faculty Sponsor)* — Associate Professor, Dept. of Pharmacotherapy & Outcomes Science; Laboratory of Pharmacogenomics & Translational Research.
- **Supervisory Committee: Franklin Bost, BPD, MBA, IDSA, FAIMBE** *(Biomedical Engineering)* · **Tamas Gal, Ph.D.** *(VCU Massey Comprehensive Cancer Center)* · **Dayanjan S. Wijesinghe, Ph.D.** *(School of Pharmacy)* · **Jonathan P. DeShazo, Ph.D.** *(Real-World Evidence Generation, CVS Health)*.

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Data Stewards, Governance & Scientific Ecosystem

- **Statewide Claims Stewardship: Virginia Health Information (VHI)** & **Virginia Center for Health Innovation (VCHI)** for Virginia APCD access under Data Use Agreement (**VCU IRB** `#HM20022300`).
- **Open Science & Peer Review: CPIC** & **U.S. FDA FAERS** pharmacovigilance resources, plus editors & peer reviewers at *Clinical and Translational Science* (ASCPT) and *Journal of Big Data*.

Personal Gratitude & Reproducible Artifacts

- **Family, Friends & Mentors:** Deepest gratitude to my family, friends, work colleagues, and mentors for their patience, encouragement, and support throughout this doctoral journey.
- **Contact & Open Artifacts: R. Jerome Dixon** (`dixonrj@vcu.edu`) · Live CDS Platform: `pgx.jerome-dixon.io` · Codebase: `github.com/Jerome3590/pgx-analysis`.

With Gratitude: Dedicated to advancing transparent, equitable, and clinician-ready pharmacogenomic decision support for high-risk patients across Virginia and beyond.

COMMITTEE Q&A BACKUP BANK

Supplemental Technical Evidence

Mathematical derivations, feature audit catalogs, pharmacogenomic scoring matrices, regulatory/HIPAA frameworks, cohort gene networks (A8), and reproducible notebook pipelines supporting defense inquiries.

🌐 Open Interactive Web Presentation (pgx.jerome-dixon.io/23NOV_defense) ↗

Appendix A1: Formal Feature Attribution (FFA) — Best-Model CNF & AXP Bounds

SPECIFIC AIM 2 · STEPS 6–8 & 10 | MATHEMATICAL VERIFICATION & PYSAT AXP

Step 7 Dual-Model Gate & Step 8 CNF Compilation

- Step 7 Dual-Model `TreeSHAP` Gate:

$$\mathcal{F}_{\text{SHAP}} = \text{supp}(\text{SHAP}_{\text{CatBoost}}) \cap \text{supp}(\text{SHAP}_{\text{BestXGB}})$$

  Requires cross-architecture agreement before symbolic extraction (since CatBoost CTR hashing precludes direct symbolic path parsing).
- Step 8 CNF Compilation on `best_xgb_variant`:
  - Parses JSON tree dumps of `best_xgb_variant` (`xgb` vs. `xgb_rf` winner by **PR-AUC** then **Recall**).
  - Explodes root-to-leaf paths into tabular intervals (`_explode_tree_to_dataframe`) and compiles Conjunctive Normal Form (CNF) clauses (`rule_clauses`).

SHAP-Guided Minimal Hitting-Set AXP & Complexity

- SHAP-Guided Rule Filter → PySAT `Hitman`:
  - Rather than brute-forcing $> 10,000$ matched tree rules per patient, prioritizes top SHAP-supported matched rules (with stratified rare-population coverage safeguards) into the PySAT `Hitman` minimal hitting-set solver to extract minimal **Abductive Explanations (AXPs / Horn clauses)** with $\text{Supp}(R) \geq 0.05$ and $\text{Conf}(R) \geq 0.70$.
- Step 8 Attribution Responsibility & Step 10 Combined Synthesis:

$$\text{AttrResp}(f) = \frac{\text{RuleFreq}(f)}{\sum_k \text{RuleFreq}(f_k)} \times \text{SHAP}_{\text{consensus}}(f)$$

$$\text{Score}_{\text{combined}}(f) = 0.5 \cdot \text{SHAP}_{\text{norm}}(f) + 0.5 \cdot \text{FFA}_{\text{norm}}(f)$$

- **Complexity vs. Neural Nets:** Tree CNF extraction is polynomial $\mathcal{O}(T \cdot L)$, whereas ReLU neural network minimal explanation is **co-NP-complete** (Katz et al., 2017).

↩ Return to Consensus Attribution (Aim 2)

Operational Threshold Validation Protocol (`SHAP ⩾ P75` ∩ `Supp ⩾ 0.05, Conf ⩾ 0.70`): Framed as a tunable *precision–volume dial* for clinical review capacity and validated via a 2-part sensitivity protocol: (1) **Two-Gate Mirror Ablation** (`SHAP-only` vs. `FFA-only` vs. `SHAP ∩ FFA` grid sweep testing top-K core signal retention, boundary noise filtering, and review-load compression); and (2) **Utilization-Free Specification Stress Test** (stripping utilization covariates/density strata to compare univariate `SHAP` Jaccard rank stability vs. symbolic `FFA` rule persistence under model perturbation).

Appendix A2: Step 3b Target Leakage — Temporal Event-Partitioning Protocol

SPECIFIC AIM 1 · STEP 3B | TEMPORAL PROCESS AUDIT

Leakage Vector Category	Representative Claims Signals	Temporal Timing	Clinical Mechanism
Post-Event OUD Rescue & MAT Induction	Buprenorphine, Suboxone, Naloxone, Methadone, Extended-Release MAT	$t \geq t_0$ (Post-Index)	Initiated in ED or bridge clinic *as a consequence following* acute overdose
Confirmatory Toxicology & Monitoring	Presumptive & definitive urine drug screens (e.g., CPT 80307, 80305, G0480)	$t \geq t_0$ (Post-Index)	Ordered during acute ED triage or post-discharge MAT compliance monitoring
Secondary Administrative Follow-Up	Long-term drug therapy status, rehabilitation & subsequent encounter billing	$t \geq t_0$ (Post-Index)	Downstream administrative claims generated directly by the index visit

Step 3b Formal Event-Partitioning Rule & Symmetric Whitelist

- **Post-Target Ratio Filter:** $\tau_{\text{post}}(f) = \frac{N(t_{\text{claim}} \geq t_0)}{N(t_{\text{claim}} < t_0) + N(t_{\text{claim}} \geq t_0)} \geq 0.80 \implies \text{Purge from } \texttt{safe_feature_filter.json}$.
- **Symmetric Pre-Index Enforcement:** All retained features satisfy $t_{\text{claim}} < t_0$ on $[t_0 - 365, t_0 - 1]$ (Opioid) or $[t_0 - 21, t_0 - 1]$ (Polypharmacy) across cases and 1:5 matched controls.

Why Unmonitored Gain Rankings Fail (`Step 3a` vs. `Step 3b`)

- **The Literature Failure (30% of PROBAST Models):**
  - Unfiltered feature selection loads top tree splits onto post-overdose MAT induction and ED toxicology screens, inflating apparent discrimination by 5–20 AUROC points.
- **Methodological vs. Empirical Integration:**
  - Architecture (`Module IV`): Locks the **Step 3b temporal event-partitioning methodology** and antecedent whitelist.
  - Empirical Results (`Module VI`): Purges **348 / 1,029 (`33.8%`)** post-index codes (`safe_feature_filter.json`), restoring honest **2019 holdout discrimination** (CH2 Results ↗).

↩ Return to Target Leakage | 📊 Return to CH2 Empirical Audit →

Appendix A3: Step 5 CPIC Drug Mapping & Patient-Level Count Features

SPECIFIC AIM 1 · STEP 5 (`5_pgx_analysis`) CLAIMS EXPOSURE VS. GENOTYPE DEFENSE

1. Global APCD Drug → CPIC Guideline Mapping (`build_global_drug_cpic_mapping.py`)

- **Official CPIC Reference Corpus:** Ingests `cpic_gene-drug_pairs.xlsx` / `cpic_drug_list.json` (`573` gene–drug pairs, `300` unique drugs, `121` pharmacogenes across CPIC Levels A–D).
- **Automated Drug Vocabulary Matching:** Extracts unique APCD drug features (`DRUG:` , `item_drug_*`) from Step 3 feature importance outputs and matches them against CPIC drug names ($\geq 95\%$ fuzzy threshold) to build `drug_cpic_mapping_global.csv` (`s3://pgxdatalake/gold/pgx_features/global/`).
- **Binary Guideline Indicator:** Determines whether or not each dispensed APCD generic medication has an official CPIC pharmacogenomic guideline ($d \in \mathcal{C}_{\text{CPIC}}$).

2. DuckDB Patient-Level Drug & CPIC Counts (`create_pgx_features_patient_level.py`)

- **Per-Patient Count Features (`mi_person_key`):** `pgx_num_drugs` $_i = |D_i|$ (total distinct pre-index drugs) and `pgx_num_cpic_drugs` $_i = |\{d \in D_i : d \in \mathcal{C}_{\text{CPIC}}\}|$ (distinct CPIC-guideline drugs).
- **Cohort Exposure Counts:** Aggregates distinct pre-index drugs (D_i) per patient in DuckDB and records cohort-level counts of patients with any drug (`pgx_num_drugs` $_i > 0$) vs. patients carrying CPIC-guideline drugs (`pgx_num_cpic_drugs` $_i > 0$).

3. Claims-Derived Exposure vs. Genotype Imputation Defense

- **Step 5 Supervised Model Pipeline (`5_pgx_analysis`):**
  - Uses **observable CPIC drug counts only** (`pgx_num_drugs` , `pgx_num_cpic_drugs`) merged via `add_pgx_features_to_model_data.py` .
  - **Claims Boundary:** Administrative claims quantify dispensed drug exposure, not molecular genotypes or germline DNA sequences.
- **Strict Exclusion of Race-as-Proxy Imputation:**
  - Administrative race codes are strictly excluded as an unscientific, inaccurate proxy for genetic ancestry or metabolizer status.
  - Preserves methodological integrity: models test whether actionable medication burden predicts ED risk without asserting unmeasured biology.
- **Translational Continuity (Aim 1 → Aim 3):**
  - Supervised models (`Steps 5-6`) evaluate observable claims CPIC exposure directly, while downstream point-of-care patient genotype & diplotype actionability cards execute in `60 ms` within our stateless Aim 3 microservice (Module VI · CH5 Results ↗).

↩ Return to Claims Join & Schema ↩ Return to Model Selection
Appendix A8: Gene Markers & Networks →

Appendix A4: Computational Pipeline & Notebook Architecture

AIM 1 AIM 2 AIM 3 FULL TRACEABILITY

Jupyter Notebook Pipeline Directory (`pgx-analysis`)

Notebook	Steps	Core Functions & Deliverables
`0_config_and_pipeline.ipynb`	Setup	Environment config, S3-to-NVMe sync, DuckDB data dictionary
`1_cohort_workflow.ipynb`	1a, 1b, 2	FAERS & APCD 21-day ADE window calibration, dual cohort extraction, 1:5 incidence-density matching, 0-day guard
`2_feature_importance.ipynb`	3a, 3b, 3c	Feature screening, Step 3b BupaR event-partitioning audit, safe antecedent whitelist
`3_model_train_shap_ffa.ipynb`	4, 5, 6, 7, 8	Density tiers, CPIC drug mapping & counts (`pgx_num_cpic_drugs`), Step 6 best-model selection, TreeSHAP & FFA rules
`3_model_sensitivity.ipynb`	CH4 Supp	Utilization-free sensitivity & claims-density stratification checks
`4_dashboard_visuals.ipynb`	9	BupaR clinical pathways, DTW trajectory clusters, FP-Growth itemsets
`5_build_and_deploy.ipynb`	10	Multi-stage Docker packaging, AWS Lambda deployment, API Gateway

Execution Model, Checkpoints & Reproducibility

- **S3 Checkpoint Idempotency:**
  - Each stage syncs inputs from `s3://pgx-analysis-data/` to local NVMe.
  - SHA-256 state markers prevent redundant compute on completed steps.
- **Dual Compute Environments:**
  - **Local Workstation:** Windows 11, DuckDB 1.4+, Python 3.11, R 4.3 (bupaR).
  - **Batch Compute:** Amazon Linux 2023 EC2 (`mushin` account `535362115856`).
- **Source Control & Public Artifacts:**
  - Monorepo: `github.com/Jerome3590/pgx-analysis`

↩ Return to Dissertation Portfolio

Appendix A5: Specific Aims, Methodological Estimators & Evaluation Metrics

SPECIFIC AIM 1 | SPECIFIC AIM 2 | SPECIFIC AIM 3 | PRE-RESULTS METHODOLOGICAL & MATHEMATICAL SPECIFICATION

Specific Aim & Pipeline Stage	Pre-Specified Design & Transformation	Mathematical Estimator & Governance Gate
Aim 1 · Steps 1–2: Matching & `0-Day` Guard	`1:5` incidence-density control sampling on `(age_band, sex, year)`; lookback $[t_0 - W,\ t_0 - 1]$	$t_{\text{claim}} < t_0 \quad (W = 365\text{d}$ `opioid_ed`$, W = 21\text{d}$ `ed_non_opioid`$)$ (purges $t = t_0$ discharge fills)
Aim 1 · Step 3b: Leakage Ratio $\tau_{\text{post}}(c)$	Event-level `bupaR` pre/post index (t_0) timing partition across candidate claims	$\tau_{\text{post}}(c) = \frac{N_{t \geq t_0}(c)}{N_{t<t_0}(c) + N_{t \geq t_0}(c)} < 0.80$ (locks symmetric `safe_feature_filter.json`)
Aim 1 · Step 4: Density Strata & Burstiness B	Training (`2016-2018`) pre-index claim quantiles ($P_{25},\ P_{50},\ P_{95}$); inter-claim gaps Δt	$b(n_{\text{ev}}) \in \{0,\ 1,\ 2,\ 3\}$ (`low` - `extreme`); $B = \frac{\sigma_{\Delta t} - \mu_{\Delta t}}{\sigma_{\Delta t} + \mu_{\Delta t}} \in [-1,\ 1]$
Aim 1 · Step 5: CPIC Drug Mapping & Counts	Global APCD drug → CPIC guideline mapping (`drug_cpic_mapping_global.csv`) & patient CPIC drug counts	`pgx_num_drugs`$_k = \lvert D_k \rvert$; `pgx_num_cpic_drugs`$_k = \lvert \{d \in D_k : d \in \mathcal{C}_{\text{CPIC}}\} \rvert$
Aim 1 · Step 6: MCCV Ensemble & Holdout	25-split stratified MCCV (`CatBoost`, `XGB`, `XGB-RF`) + Platt scaling on untouched `2019` holdout	$\hat{p}_{\text{ens}}(\mathbf{x}) = \sum_{m=1}^{3} w_m \hat{p}_m(\mathbf{x}),\quad w_m = \frac{\text{PR-AUC}_m}{\sum_j \text{PR-AUC}_j}$; $\text{Lift} = \frac{\text{AUPRC}}{\pi},\quad \text{BS} = \frac{1}{N}\sum (\hat{p}_k - y_k)^2,\quad \text{ICI}$

Specific Aim & Attribution Stage	Pre-Specified Design & Demarcation	Mathematical Estimator & Governance Gate
Aim 2 · Steps 7–8: Consensus Gate $\mathcal{C}$ ★	Dual-model `TreeSHAP` (`Cat` ∩ `Best XGB`) + PySAT `Hitman(m22)` minimal hitting-set `AXP`	$\overline{\lvert\phi_j\rvert} \geq Q_{75} \cap \text{Supp}_{\text{AXP}}(f_j) \geq 0.05$ $\cap\ \text{Conf}(r) = \frac{N(r \wedge \hat{y}=1)}{N(r)} \geq 0.70$
Aim 2 · Step 8: Intervention Rate (`IR`) ★	In-model consensus risk sensitivity contrast ($f_i \to x_{\text{med}}$, non-causal); never from FP-Growth	$\text{IR}_{\Delta\hat{p}}(f_i) = \frac{1}{n}\sum_{k=1}^{n}\left(\hat{p}^{(k)} - \hat{p}^{(k)}_{f_i \to \text{med}}\right)$; $\text{IR}_{\text{AXP}}(f_i) = \frac{1}{\lvert E_1 \rvert}\sum_{k \in E_1} \mathbb{I}\left(\hat{y}_{f_i \to \text{med}} = 0\right)$
Aim 2 · Step 8: Interaction Effect (`IE`) ★	In-model consensus risk contrast Δ_{11} & Class-1 `AXP` co-occurrence lift (non-causal; 1,000 bootstrap CIs)	$\Delta_{11} = \mathbb{E}[\hat{p}_{11}] - \mathbb{E}[\hat{p}_{10}] - \mathbb{E}[\hat{p}_{01}] + \mathbb{E}[\hat{p}_{00}]$; $\text{IE}(f_i, f_j) = \frac{P(f_i \wedge f_j \mid \text{AXP}_1)}{P(f_i \mid \text{AXP}_1)\,P(f_j \mid \text{AXP}_1)} > 1.0$
Aim 3 · Step 9: Visual Analytics Demarcation	Training-cohort `bupaR`, `DTW`, & `FP-Growth`; **strictly excluded from predictor matrix**	$D_{\text{DTW}} = \min_{\pi \in \mathcal{A}_w} \sqrt{\sum \lVert \mathbf{x}_{i,t_1} - \mathbf{x}_{j,t_2} \rVert_2^2}$; $\text{Lift}_{\text{FPG}}(A \to B) = \frac{\text{Supp}(A \cup B)}{\text{Supp}(A)\,\text{Supp}(B)}$
Aim 3 · Step 10: OODA Scenario Delta $\Delta\hat{p}$	Stateless consensus risk scenario delta ($\mathbf{x} \to \mathbf{x}'$); Imputation of Normality ($x_{\text{unobs}} = 0$)	$\Delta\hat{p} = \hat{p}_{\text{ens}}(\mathbf{x}') - \hat{p}_{\text{ens}}(\mathbf{x})$; $\text{Score}_{\text{comb}}(f) = 0.5 \cdot \text{SHAP}_{\text{norm}}(f) + 0.5 \cdot \text{FFA}_{\text{norm}}(f)$

Methodological-to-Empirical Traceability: Locks pre-specified study designs, mathematical estimators, and strict predictive-vs.-visual demarcation across Specific Aims 1–3, directly governing the 2019 temporal holdout outcomes, `IR` / `IE` rankings, and sensitivity stress tests presented in **Module VI (Results)** .

↩ Return to Consensus Attribution (Aim 2) | ↩ Return to Methodological Fences | Appendix A6: Master Acronym Reference →

Appendix A6: Master Acronym & Technical Nomenclature Reference

GOVERNANCE: CRISP-DM · TRIPOD+AI · OODA | APCD · HCG (MILLIMAN) · CPIC PGX | GBDT · TREESHAP · PYSAT FFA/AXP | SERVERLESS CDS & MTM

Acronym	Full Expansion	Operational Role in Dissertation
`CRISP-DM`	Cross-Industry Standard Process for Data Mining	6-phase systems engineering lifecycle governing Steps 1a–10 (Business Understanding → Data Prep → GBDT Modeling → Lambda Deployment).
`TRIPOD+AI`	Transparent Reporting of a Multivariable Prediction Model (+ AI)	27-item biomedical validation standard enforcing Type 2b/3 temporal split (`2016-2018` train → untouched `2019` holdout; zero recalibration).
`OODA` Loop	Observe–Orient–Decide–Act Loop	Clinical workflow boundary: GBDT & Horn rules *Observe/Orient*; human pharmacists retain 100% authority to *Decide/Act*.
`PROBAST` / `SQLR`	Prediction Model Risk of Bias Assessment Tool / Systematic Quant. Lit. Review	Chapter 1 PRISMA 2020 review ($n = 94$ ML studies; $n = 73$ PROBAST models) exposing the **"Empty Intersection"** ($n = 0/73$).
`APCD` / `VHI`	All-Payer Claims Database / Virginia Health Information	Statewide longitudinal medical & adjudicated pharmacy claims lake (`2016-2019` , 4.2M patients, 380M claims, deterministic `mi_person_key`).
`HCG` (Milliman)	Milliman Health Cost Guidelines (HCG Grouper)	Standardized APCD encounter acuity taxonomy (`hcg_setting` , `hcg_line` , `hcg_detail`) anchoring ED visits: `P51` / `P51b` (PHY ED), `O11` (ER), `P33` (Urgent Care).
`CPIC` / `PGx`	Clinical Pharmacogenetics Implementation Consortium / Pharmacogenomics	Step 5 maps APCD drugs to CPIC guideline status (`drug_cpic_mapping_global.csv`) to quantify patient-level CPIC medication exposure burden (`pgx_num_drugs` , `pgx_num_cpic_drugs`).
`ADE` / `DDI` / `FAERS`	Adverse Drug Event / Drug–Drug Interaction / FDA Adverse Event Reporting System	Geriatric **"Silent ADEs"** (`R29.6` falls, `S72` hip fx, `R55` syncope, `N17` AKI) & FDA FAERS 21-day onset kinetics ($> 90\%$ acute ADEs).
`OUD` / `MME` / `MAT`	Opioid Use Disorder / Morphine Milligram Equivalents / Meds for Addiction Tx	12-month chronic opioid trajectory anchors (`F11.20` , `T40.0X-T40.4X`) vs. post-index rescue drugs purged by Step 3b (buprenorphine, naloxone).

Acronym	Full Expansion	Operational Role in Dissertation
`GBDT` / `XGB-RF`	Gradient-Boosted Decision Trees / XGBoost Random Forest	Piecewise-constant ensembles comparing oblivious trees (`CatBoost`), depth-wise trees (`XGBoost`), and bagged trees (`XGBoost-RF`).
`MCCV` / `TPE`	Monte Carlo Cross-Validation / Tree-structured Parzen Estimator	25-split stratified development CV & Bayesian hyperparameter optimization (`Optuna` , 50 trials) with Platt probability calibration.
`AUROC` / `PR-AUC` / `ICI`	Area Under ROC / Precision-Recall AUC / Integrated Calibration Index	Discrimination, imbalanced minority-class precision-recall lift, Brier Score, ICI calibration, and Decision Curve Analysis (`DCA`).
`XAI` / `TreeSHAP`	Explainable AI / Tree SHapley Additive exPlanations	Exact game-theoretic feature attribution (ϕ_j) intersected across `CatBoost` ∩ `Best XGBoost` ($\overline{\lvert\phi_j\rvert} \geq Q_{75}$) in Step 7.
`FFA` / `CNF` / `PySAT`	Formal Feature Attribution / Conjunctive Normal Form / Python SAT Solver	Step 8 compilation of tree paths into CNF clauses solved via PySAT `Hitman` minimal hitting sets in polynomial $\mathcal{O}(T \cdot L)$ time.
`AXP` & Horn Rules	Abductive Explanation & Boolean Horn Logic Clauses	Minimal irreducible `IF-THEN` conditions ($P_1 \wedge P_2 \implies \hat{y} = 1$) guaranteeing Class-1 risk ($\text{Supp} \geq 0.05, \text{Conf} \geq 0.70$).
`IR` & `IE` (Δ_{11})	Intervention Rate & Interaction Effect (2nd-Order Risk Contrast)	In-model consensus risk attenuation drop ($\text{IR}_{\Delta\hat{p}}, \text{IR}_{\text{AXP}}$) and cross-partial synergy lift ($\Delta_{11}, \text{IE} > 1.0$) strictly from trees/AXPs.
`bupaR` / `DTW` / `FPG`	Business Process Analysis in R / Dynamic Time Warping / FP-Growth	Exploratory process flows, trajectory distance (D_{DTW}), and itemset lift (Lift_{FPG}) strictly excluded from supervised GBDT matrices.
`CDS` / `MTM` / `PHI`	Clinical Decision Support / Medication Therapy Mgmt / Protected Health Info	Aim 3 stateless, zero-PHI AWS Lambda microservice ($\mu_{\text{warm}} = 6\text{ms}, < \$5/\text{mo}$) driving an asynchronous Tier 1–3 pharmacist review queue.

↩ Return to Governance and Frameworks | ↩ Return to Polypharmacy Cohort Pipeline | ↩ Return to Methodological Fences | Appendix A7: Regulatory & HIPAA Scope →

Appendix A7: Regulatory Scope Demarcation & HIPAA Cloud Architecture

SPECIFIC AIM 3 · CLINICAL DEPLOYMENT & TRANSLATION | REGULATORY SCOPE: FTC ACT VS. HIPAA BAA | AWS HIPAA-ELIGIBLE ENCLAVE & SHARED RESPONSIBILITY

1. Scope Demarcation: Direct Patient vs. Hospital Deployment

- **Direct Patient Self-Use (FTC Consumer Scope):**
  - Direct individual input of personal health or genetic data is governed by **FTC Act Section 5** and the **FTC Health Breach Notification Rule (HBNR)**.
  - HIPAA applies *exclusively* to covered entities and business associates; when a consumer directly uses the tool, a BAA is **neither required nor legally valid**.
- **Hospital / Healthcare Organization Use (HIPAA Scope):**
  - Clinical deployment processing patient records constitutes ePHI under **45 CFR § 160.103**.
  - **Two-Tier Legal BAA Execution Chain:**

  - Entity executes BAA with hospital; entity executes BAA with AWS via AWS Artifact.
- **VCU CCTR Pilot Grant Integration:**
  - VCU Wright Center CCTR $25,000 Grant formally funds Aim 2 Compliance & Legal Review ($4,500) to partner with VCU Health Compliance and Counsel to execute institutional DUAs and BAAs prior to prospective clinical pilot testing.

2. AWS Technical Reality: "HIPAA-Eligible" vs. "HIPAA-Compliant"

- **AWS Shared Responsibility Model:**
  - AWS services are *HIPAA-eligible*; application architecture must implement mandatory administrative, physical, and technical safeguards.
- **Mandatory Cloud Technical Configurations:**
  - **S3 Buckets:** Enforce AWS KMS encryption (`SSE-KMS`), block all public access, enforce TLS (`aws:SecureTransport`) in bucket policies.
  - **AWS Lambda & Memory Security:** Environment variables KMS-encrypted; Firecracker microVM `/tmp` wiped post-invocation; zero ePHI persisted.
  - **CloudWatch Log Scrubbing (Critical):** Custom log sanitizers strictly prevent ePHI from leaking into log streams, stdout, or telemetry.
  - **CloudTrail Auditing:** Immutable management and data API event trails under **45 CFR § 164.312(b)**.
- **AI & External API Boundaries:**
  - **AWS-Native:** Amazon Bedrock / SageMaker models hosted within VPC fall under AWS BAA.
  - **External Third-Party APIs (OpenAI, Anthropic):** Inbound/outbound calls require a separate signed BAA with each vendor, regardless of ephemeral processing.

↩ Return to Aim 3 Decision Support (Slide 20) | ↩ Return to Methodological Fences (Slide 22)
Appendix A8: Gene Markers & Networks → | Technical Appendix Overview →

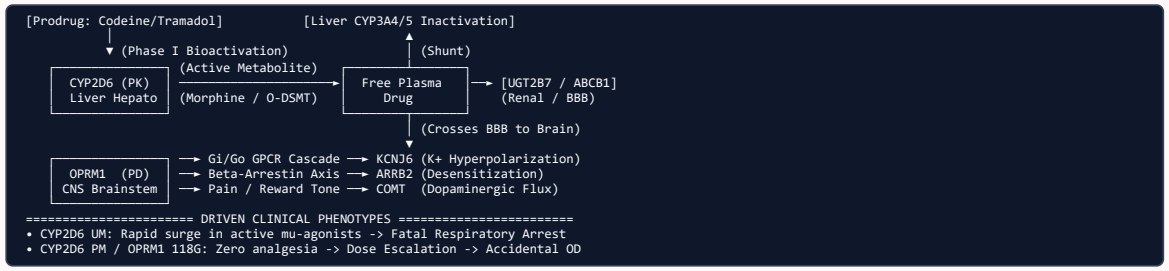
Architectural Synthesis: Ephemeral compute with zero persistent PHI is a sound engineering safeguard, but enterprise clinical deployment legally requires the **two-tier BAA chain** and AWS HIPAA hardening. VCU CCTR Pilot Grant funding guarantees institutional legal review and DUA execution before prospective clinical pilot deployment.

Appendix A8: Pharmacogenomic Gene Markers & Mechanistic Networks by Cohort

SPECIFIC AIM 1 · CPIC & PHARMGKB VIP MECHANISTIC EVIDENCE | OPIOID COHORT: CYP2D6, OPRM1 | POLYPHARMACY COHORT: CYP2C19, CYP2C9, SLCO1B1, VKORC1

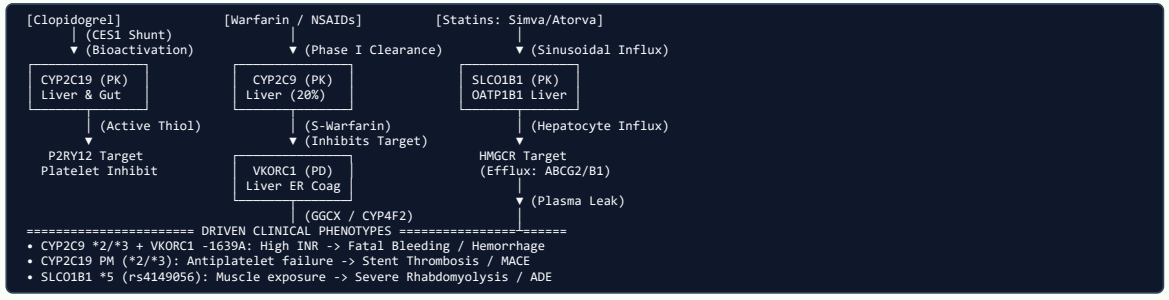
1. Opioid Cohort: CYP2D6 (PK) & OPRM1 (PD) Network

- **Organs & Pathways: Liver** (hepatic phase I bioactivation of Codeine → Morphine, Tramadol → O-DSMT) & **Brain/CNS** (μ-opioid receptor G_i/G_o GPCR cascade in brainstem pre-Bötzinger respiratory pacemakers, PAG, and NAc reward circuits).
- **Supporting Network:** `CYP3A4/5` (inactivation shunt), `UGT2B7` (glucuronidation clearance), `ABCB1` (BBB P-gp efflux), `ARRB2/ARR3` (β-arrestin desensitization), `KCNJ6` (GIRK2 K^+ channels), `COMT` (pain/reward valence).
- **Driven Phenotypes: CYP2D6 UM** causes toxic active surge → *Fatal Respiratory Depression*; **CYP2D6 PM** (`*3/*4/*5`) & **OPRM1 118G** (`rs1799971`) cause non-activation/blunted binding → *Analgesic Failure & Dose-Escalation Overdose*.



2. Polypharmacy Cohort: CYP2C19, CYP2C9, SLCO1B1, VKORC1

- **Organs & Pathways: Liver** (sinusoidal uptake, phase I clearance, ER coagulation synthesis), **Vascular Endothelium** (platelet $P2Y_{12}$ inhibition), & **Skeletal Muscle** (statin exposure target).
- **Supporting Network:** `CES1` (85% clopidogrel hydrolysis), `P2RY12` (platelet receptor), `CYP4F2` (vitamin K_1 clearance), `GGCX` (γ-carboxylase), `HMGCR` (statin target), `ABCG2/ABCB1` (efflux transporters).
- **Driven Phenotypes: CYP2C9 (`*2/*3`) + VKORC1 (`-1639A/A`)** → *Supratherapeutic INR & Fatal Bleeding*; **CYP2C19 PM (`*2/*3`)** → *Clopidogrel Failure & Stent Thrombosis/MACE*; **SLCO1B1 (`*5` rs4149056)** → *Severe Myopathy & Rhabdomyolysis*.



↩ Return to CPIC FI (Slide 9) | ↩ Return to Step 5 (A3) | Appendix Bank →

Mechanistic Synthesis: Claims-derived CPIC exposure signals reflect profound pharmacogenomic biological vulnerabilities. In `opioid_ed`, divergent `CYP2D6` clearance and `OPRM1` signaling drive respiratory arrest versus undertreatment-induced overdose. In `non_opioid_ed`, high-acuity elderly ADEs stem from dual PK/PD coagulation failure (`CYP2C9` / `VKORC1`), antiplatelet resistance (`CYP2C19`), and statin transport failure (`SLCO1B1`).